P425/2

PRINCIPAL MATHEMATICS

PAPER 2

July/Aug 2026

3 Hours

ASSHU ANKOLE JOINT MOCK EXAMINATIONS 2026

Uganda Advanced Certificate of Education

PRINCIPAL MATHEMATICS

P425/2

Time: 3 Hours

Instructions:

*This paper consists of **two** sections;* **A** and **B.**

*It has **five** examination items.*

*Section **A** has **one** compulsory item.*

*Section **B** has two parts; I and II. Respond to **only one item** from **each part.***

*Respond to **three items** in all.*

*Any additional **item(s)** responded will **not** be scored.*

*All responses **must** be written in the answer booklet(s) provided.*

*All responses **must** be shown **clearly.***

Responses to each item should start on a fresh page.

Indicate the item number you are responding to, clearly at the top of the page.

Silent, non-programmable scientific calculators and mathematical tables with a list of formulae may be used.

Acceleration due to gravity ($g = 9.8ms^{-2}$)

SECTION A (NUMERICAL METHODS)

(COMPULSORY ITEM)

ITEM ONE

National Water and Sewerage Cooperation (NWSC) is encountered with a problem of the volume capacity of their cylindrical pipes that supply water to their customers. They consulted their engineer who found out that the diameter is 0.62metres and length 6.2metres. There was need to know the range between which the volume of cylindrical pipe lie, the engineer advised the NWSC manager that for the cylindrical pipes to provide sufficient water supply to its customers, the range should be between 1.6 and 2.0.

The NWSC was also encountered with a problem of location of a point at which to locate a pumping machine along a linear bank of a river that stretches from 3km to 5km. Previously data collected by the engineering department had indicated that this area was bounded by a line($y = 4x + 6$) and another curve ($y = 3e^{0.5x}$). The surveying department noticed that using a graphical method the point of intersection between the above two curves that lie along the line from 3km to 5km above would be a suitable approximate point for the location of the pumping machine and required that it should be rounded off to 1 decimal place, use a suitable iterative method to get the exact value, if this point lies between 3.5 and 4, it would be a suitable approximate point at which to install the pumping machine.

The NWSC was also encountered with a problem of establishing the costs and distances involved in transportaticn of their water installation materials. It had been noted for distances of 30km, 40km, and 50km the costs were sh. 60,000/=, sh. 100,000/= and sh. 130,000/= respectively. They were approached by two of their customers, one paid sh. 150,000/= and another one was to be left at a distance of 46km. There was need to use a suitable linear method to estimate for them the distance and the cost. The second customer complained that he had been overcharged by paying sh. 120,000/= and there was need to justify his claim.

TASK:

As a mathematics student, analyze the problems of NWSC by giving an interpretation, methodology and justification.

SECTION B: PART I (PROBABILITY AND STATISTICS)

(Attempt one item from this part of the section)

ITEM TWO

An entrepreneur has two companies producing two types of beverages Pepsi Cola and Mountain Dew. He consults his business analyst over his companies. The business analyst finds out that there is one company located in Apac (A) and another in Bukedia (B). The data analyst finds out that the likelihood of customers buying from district A and B are in the ratio 2:1 and that the ratio in thousands produced by company A are 6 Pepsi Cola and 8 Mountain Dew and that produced in

company B are in the ratio of 9 Pepsi Cola and 7 Mountain Dew. The business analyst informed the entrepreneur that there was need to make 2 pickings and find out the probability that the beverages are of the same type and if it is above 0.4 then the companies would increase the production of beverages of the same type. He further noticed that there was need to get the probability that they are from A given that they are of the same type.

The business analysis advised that there was need to develop a probability distribution of Mountain Dew type and find out the mean of the distribution of Mountain Dew and if it is above 1, then the Mountain Dew section would employ more people. He advised there was need also to find the standard deviation of the distribution.

The data analyst found out that there were other 30 similar companies countrywide and taking the probability of success is that of the Bukedia Company, he advised that there was need to get the probability of at least 9 successful companies and if it is above 0.2, then there would be a bank opened specifically for the Bukedia Company.

TASK:

As a Mathematics student, interpret, show the methodology and make a valid justification.

ITEM THREE

The board of governors (B.0.G) of a school is encountered with a problem of analysis of scores of their students during the end of term one that were compiled by the Director of Studies (D.O.S) of their school that was as indicated below.

Scores	5-<30	30-<40	40-<50	50-<60	60-<80
Number of students	17	21	30	14	3

The board of governors contacted a statistician who advised them that there was need of a graphical representation to help in knowing the mode, he also advised that a calculation for average score and standard deviation was necessary. He advised the board of governors that if students on average scored an average score less than 45 then all parents would be called for an academic assembly.

When the data was analyzed by the statistician, it was found out that it was approximately normally distributed. He advised the board of governors that there was need to know the 95% confidence interval around the average score. And if it lies between 35 and 45, there would be no need to administer remedial tests.

The statistician selected a sample of 10 students, where the probability between 35 and 45 taken as a normal distribution would be considered as a probability of success and therefore there was need to find the probability that at most 2 students are successful. And if the probability was found to be less than 0.2 then the director of studies would administer a suitable punishment.

TASK

As a mathematics student, interprete, show the methodology and give justification to the board of governors.

SECTION B: PART II (MECHANICS)

(Attempt one item from this part of the section)

ITEM FOUR

A business man wants to install a regular pentagon solar panel structure, he consults the engineer who finds out that the regular pentagon solar panel structure is of mass 10kg and length 2m with forces of 6N, 13N, 5N, 2N and 3N acting along the sides AB, BC, CD, DE and AE respectively, the directions of the forces are in the order of the letters. The engineer took AB as the reference horizontal axis. The engineer advised that there was need to know the resultant force and should not be beyond 18N for the cost of the solar panel (ABCDE) to be affordable and it was important to know the direction of the resultant force.

The engineer informed the business man that the regular pentagon solar panel was installed at the top of the roof at a distance 4m from the edge of roof that is inclined at 30° to the horizontal. There was need to know velocity with which it reaches the edge in case it is detached from the roof and slides down a rough with coefficient of friction of a $\frac{1}{4}$, he advised that the velocity should not go beyond $10ms^{-1}$ otherwise there will be loss of energy.

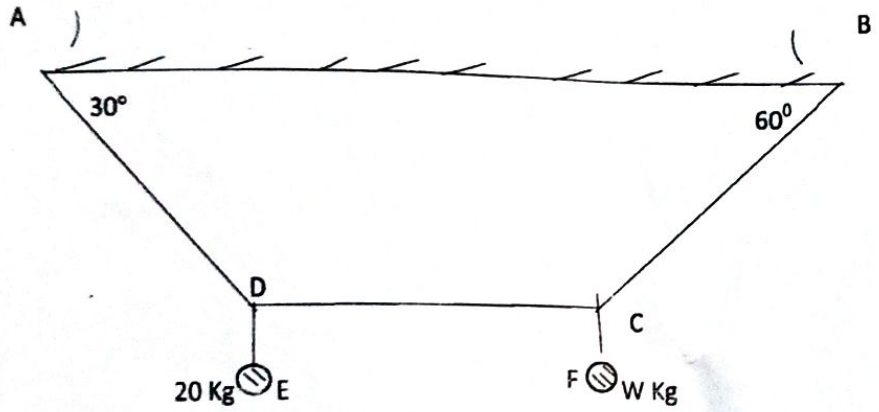
The engineer noted that regular pentagon solar panel was projected below the horizontal at an angle of 30^0below the horizontal with a velocity above. The height above the ground to the edge was measured as 4m by the engineer. There was need to know the horizontal distance that is from the height to where the rectangular solar panel hits the ground. He preferred that it does not go beyond 3m since there are usually children who are playing. There is need to know the safety of these children.

TASK

As a mathematics student, analyse and justify the regular pentagon solar panels problems to the business man.

ITEM FIVE

A hotel management has a lighting system that it is being investigated about, it consults its engineer who informs them that the lighting system is attached to three strings AD, DC and BC. The stringsAD and BC are fixed to points A and B. Bulbs of mass 10kg and W kg are attached at points D and C respectively and rest in equilibrium position as shown below.



The engineer advised the hotel management that there was need to dctermine the tensions in the strings AD, DC, BC and weight, if W exceeds 15kg there is likelihood that the string will break.

The engineer observed that the string broke and the particle at F slid from rest down a rough inclined plane that is at an angle of 30° to the horizontal which had a co-efficient of friction of $\frac{1}{5}$, it slide for a distance of 6m. He advised the hotel management that there was need to know the velocity on reaching the lowest point of the plane and if it is less than $15ms^{-1}$ there would be no destruction of the bulb.

The engineer observed that on reaching the ground level, it slid on a rough horizontal plane which had a co-efficient of friction of $\frac{1}{4}$ with a force of 130N and moved until it came to rest. There is a wall at a distance of 10m and the engineer advised that there was need to know the safety of the wall.

TASK

As a mathematics student, help the hotel management to interpret, use a suitable methodology and give justification on the findings of the engineer.

END

UGANDA ADVANCED CERTIFICTE OF EDUCTION

S.6 PRE MOCK EXAMINATION - SET ONE

PRINCIPAL MATHEMATICS

PAPER 2

Time: **2HOURS 30 MINUTES**

INSTRUCTIONS TO CANDIDATES:

- *This paper consists of* **two** *sections;* **A** and **B.** *It has five examinations items.*
- Section **A** has **one compulsory** item.
- *Section **B** has two parts; **part I** and **part II**. Answer one item from each part.*
- *Answer a total of* **three** *items.*
- *Any additional item(s) answered will* **not** *be scored.*
- **All** *answers* **must** *be written in the answer booklets/sheets provided.*
- *Graph paper is provided.*
- *Silent non-programmable scientific calculators and mathematical tables with a list of formulae may be used.*

<table>
<tr><td></td><td colspan="3">FOR OFFICIAL USE ONLY</td></tr>
<tr><td>SECTION A</td><td colspan="2">ITEM 1</td><td></td></tr>
<tr><td rowspan="4">SECTION B</td><td rowspan="2">PART 1</td><td>ITEM 2</td><td></td></tr>
<tr><td>ITEM 3</td><td></td></tr>
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<tr><td>ITEM 5</td><td></td></tr>
</table>

SECTION A
This item is compulsory

ITEM 1

A district water authority is installing an automated monitoring system at a newly constructed water treatment plant in Mbarara City. During testing, engineers discovered that the sensor controlling the water flow rate operates correctly only when a parameter, (x), satisfies the equation

$x^2 - \cos x = 0$.

To avoid delaying the commissioning of the plant, a trainee engineer suggests first determining the interval containing the positive root of the equation and then using one application of linear interpolation to obtain an approximate value of (x). However, the chief engineer recommends refining this estimate using the Newton–Raphson method before approving the sensor for operation.

According to the plant's quality assurance guidelines, the sensor may only be commissioned if the percentage error in the estimated operating value does not exceed 0.5% when compared with the accepted value of 0.824.

Task

(a) Assist the trainee engineer to determine the interval in which the operating value of (x) lies and obtain an approximate value of (x).

(b) Assess whether the sensor should be commissioned.

SECTION B
Part I
Attempt only ONE item from this section

ITEM 2

A mathematics club is preparing a probability challenge for visitors during the school's Open Day. Three games have been designed, and the patron intends to include only games that are sufficiently challenging but fair.

In the first game, a contestant tosses a biased coin five times. A head is three times as likely to occur as a tail. The game is considered suitable if the probability of obtaining at most two tails lies between 0.80 and 0.95.

Contestants who complete the first game proceed to a second game involving three identical boxes A, B, and C.

Box	A	B	C
Red balls	4	6	3
Green balls	2	7	5

One of the boxes is chosen at random and two balls are drawn successively without replacement. Let X denote the number of green balls obtained. The patron considers the game suitable if the variance of X is less than 0.5, indicating that the outcomes are reasonably predictable.

Contestants who reach the final stage spin a pointer whose score is represented by a continuous random variable Yhaving probability density function

$$f(y) = \begin{cases} \frac{2}{13}(y+1), & 0 \le y \le a, \\ \frac{2}{13}(5-y), & a \le y \le b, \\ 0, & \text{otherwise.} \end{cases}$$

The patron wishes most contestants to score below 2.5points in the final stage and therefore requires that $P(Y < 2.5)$exceeds 0.5.

Task

(a) Determine whether each of the three games satisfies the patron's requirements.

(b) Hence, advise the patron whether all three games should be included in the probability challenge.

ITEM 3

The headteacher of a secondary school in Ntungamo District is investigating whether performance in Fine Art can be used to identify students who may require additional support in Mathematics.

A sample of ten Senior Six students was selected and their marks in Fine Art $(x)$and Mathematics (y)were recorded as shown below.

Fine Art (x)	80	76	96	41	68	31	42	88	68	91
Mathematics (y)	43	32	27	64	65	64	65	32	64	43

Rebecca sat the same examinations and scored 61% in Mathematics, but her Fine Art mark was not recorded.

Records from previous years indicate that Mathematics marks in Senior Six are normally distributed with mean 48% and standard deviation 3%. A student who scores below 40% in Mathematics is regarded as having failed.

The headteacher plans to organise remedial lessons for students who are likely to fail. Each remedial group requires one box of chalk. A supplier delivers boxes containing 12 pieces of chalk, of which 3 are usually broken. To assess the reliability of the supplier, the school inspects 40 boxes from a consignment. The supplier will be retained only if the probability that between 8 and 13 of the inspected boxes contain broken pieces exceeds 0.70.

The school management committee also wishes to examine whether differences in family size within the community are large enough to be considered when planning learner-support programmes. Data collected from 100 families in a nearby village are shown below.

Number of children	1	2	3	4	5	6	7	8
Number of families	8	9	16	25	20	12	6	4

The committee agreed that family-size variation would be considered substantial if the standard deviation exceeds 2.

Task

(a) Analyse the relationship between Fine Art and Mathematics performance and estimate Rebecca's Fine Art mark.

(b) Determine the probability that a randomly selected student passes Mathematics.

(c) Hence, advise the headteacher whether:

- Fine Art performance can be used as a useful indicator of Mathematics performance;
- the supplier should be retained; and

- family-size variation is substantial enough to be considered when planning learner-support programmes.

Part II

Attempt only ONE item from this section

ITEM 4

During a community sports festival in Kabale District, Sarah stands on top of a hill 100 m above a level playground and throws a stone at an angle of $30°$ below the horizontal. The stone lands on the playground 25 m from the foot of the hill and then slides along the rough ground before coming to rest. The coefficient of friction between the stone and the ground is 0.35.

Meanwhile, Moses and Amina are operating two camera drones to record the event. At a particular instant, Moses' drone A is 8.0 km due north of Amina's drone B. Drone A flies due west at a constant speed of 150 km h^{-1}, while drone B flies at a constant speed of 200 km h^{-1}in the direction $N30°W$.

The organisers intend to use similar arrangements during a national athletics competition. However, they will proceed only if:

- the stone travels less than 15 m after striking the playground;
- the angle at which it strikes the playground does not exceed $75°$to the horizontal;
- the drones remain at least 2.0 km apart throughout the recording.

Take $g = 9.8$ m s^{-2}.

Task

(a) Determine the quantities required to evaluate the organisers' conditions.

(b) Hence, justify whether the organisers should use similar arrangements during the national athletics competition.

ITEM 5

Kato is preparing exhibits for a science fair. One exhibit consists of a square metal frame $ABCD$of side $a$units. Forces of magnitudes $4N$, $5N$, $5N$, $4N$and $6N$ act along AB, BC, CD, DA and AC respectively, in the directions indicated by the order of the letters.

The exhibit is connected to a $2kg$ block resting on a rough plane inclined at $30°$to the horizontal. The coefficient of friction between the block and the plane is 0.25. Kato intends to hold the block in equilibrium by applying a force acting up the plane at an angle of $30°$to the line of greatest slope.

A second exhibit consists of a particle P of mass $8kg$ resting on a smooth horizontal table and attached by light inextensible strings to particles Q and R of masses $2kg$ and $6kg$ respectively. The strings pass over smooth pulleys at opposite edges of the table so that Q and R hang freely. The system is released from rest. After descending $1m$, particle R strikes an inelastic floor and comes to rest.

Kato has only one support cable available. He will use it for the block on the inclined plane provided that the least force required to keep the block in equilibrium does not exceed the magnitude of the resultant force acting on the square frame. He will also use the second exhibit only if particle Q rises less than $1.5m$ before momentarily coming to rest.

Take $g = 9.8ms^{-2}$.

Task

Using appropriate calculations, determine whether Kato's arrangements will work as planned.

END

P425/2
PRINCIPAL MATHEMATICS
PAPER 2
July/August, 2026
3 hours

TV MATHSHUB MOCK EXAMINATIONS ,2026

Uganda Advanced certificate of Education

PRINCIPAL MATHEMATICS

Paper 2

3 hours

INSTRUCTIONS TO CANDIDATES

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- ***Graph paper is provided.***
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TV MATHSHUB

SECTION A

*The item in this section is **compulsory***

Item 1

A municipality is planning to improve the water supply system in its administrative centre. Since the existing water sources are insufficient, the municipal authorities have decided to obtain additional water from a nearby river which has a uniform width of exactly $20m$. The Municipal Water Engineer has been assigned the responsibility of determining a suitable location for installing a water intake pump along the river.

To analyse the river conditions, the engineer represents the depth of the river by the mathematical model: $y(x) = 12 - 0.12(x - 10)^2; \quad 0 \leq x \leq 20$

where x represents the distance in metres measured from one river bank.

The engineer states that the pumping system can only operate effectively if the cross-sectional area of the river is at least $145m^2$. According to his calculations, the given model meets this required minimum area.

During a meeting with the municipal authorities, the engineer explains that by taking depth measurements at intervals of exactly $5m$ from the river bank and applying an approximate integration method, he can verify that the river model gives the required minimum cross-sectional area. He further explains that the same condition can also be confirmed using an exact integration method.For the pump foundation to be constructed safely, the engineer requires the river depth at the installation point to satisfy: $y(x) = 8m$

He informs the authorities that by taking $4m$ from the river bank as an initial estimate, he can determine the required position of the pump using an iterative method. The position obtained should have an error tolerance not exceeding $0.1m$ to allow a stable installation.

The municipal authorities require evidence that the proposed river model satisfies the minimum area requirement using both the approximate and exact approaches. They also require the precise location where the pump should be installed to achieve the required depth.

Task:

(a) Provide proof need by the municipal authorities.

(b) Determine and advise the municipal authorities on the exact distance from the river bank where the pump should be installed.

SECTION B

*Answer only two items from this section, choosing **one item** from each of the parts I and II.*

PART I

Item 2

James is a university engineering graduate who, due to employment challenges, took a part-time teaching job at his former high school. During a mid-term assessment, he administered a test to his S.5 Applied Mathematics class. The scores obtained by the 200 learners are summarized below:

Marks	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90
Frequency	18	34	58	42	24	10	6	8

James wants to determine if his class performance has improved compared to the previous year, when 150 out of 250 students failed to obtain the pass score of 40.

Later, seeking higher income to sustain his family, Mutima transitioned from teaching to establish a commercial bakery. He acquired a packing machine to automatically seal loaves of bread. The weight of the loaves produced by his bakery follows a normal distribution with a mean weight of 5.0 kg and a standard deviation of 0.2 kg.

During initial trials, some loaves had either too little or too much dough, leading to packing machine errors. On the first day of production, 1.5% of the loaves were rejected for being underweight and 1.0% were rejected for being overweight.

TASK

(a) Is there an improvement in students' grades, if pass mark is fixed at 40%?

(b) Find the acceptable weight range for the loaves of bread to ensure they are not rejected.

(c) If Mutima adjusts the automated machinery so that only 0.5% of the loaves are rejected for being overweight, determine the new maximum acceptable weight threshold.

Item 3

In 2024, the Auditor General released a report indicating that several newly constructed major roads in Uganda failed to meet minimum quality standards, deteriorating rapidly before their projected lifespan. In response, the Ministry of Works and Transport initiated a strict quality assessment framework.

During the rehabilitation of the Matugga–Kasangati road, civil engineers collected soil samples at different depths to analyze the relationship between the soil depth and its sand composition. The percentage of sand, y, at various soil depths, x(in cm), was recorded as follows:

x	35	65	55	25	45	75	20	90	51	60
y	86	70	84	92	79	68	96	58	86	77

Simultaneously, the Ministry contracted an engineering firm to audit the quality of steel bolts used in bridge construction. The firm's quality assurance supervisor

monitors a metal workshop where these bolts are manufactured. Each day, the supervisor randomly samples 50 bolts and records the number of defective bolts,X, where $15 \leq X \leq 20$.

From historical inspection records, the supervisor identified a clear pattern in the daily defects:

- For $x = 15, 16, 17$, the probability of getting x defective bolts is proportional to $(x - 12)$.
- For $x = 18, 19, 20$, the probability is proportional to $(25 - x)$.

The workshop records show that the probability of the number of defective bolts exceeding 18 is $\frac{11}{54}$ The plant manager wishes to determine whether to recruit more inspectors or replace the manufacturing machines based on the following operational criteria:

Recruitment Policy: If the expected number of defective bolts exceeds 18.5, more inspectors must be recruited. Replacement Policy: If the probability of getting between 16 and 19 defective bolts (inclusive) exceeds 0.4, and the variance of the defects exceeds $\sqrt{5}$, then the manufacturing machine must be replaced.

Task:

(a) Comment on the relationship between the depth of soil and the percentage of sand in the soil.

(b) Advise the company whether to:

  (a) recruit more inspectors, or

  (b) replace the machines.

PART 2

Item 4

Due to unstable security conditions along a border region, the Ministry of Defense has established a joint command center to manage both airspace defense and humanitarian logistics. During an active defensive patrol, radar systems track two aircraft. To the pilot of a reconnaissance bomber aircraft flying with a true velocity $\begin{pmatrix} -150 \\ -200 \end{pmatrix}$ kmh^{-1}, an encroaching unidentified fighter aircraft appears to have a relative velocity $\begin{pmatrix} 150 \\ 440 \end{pmatrix}$ kmh^{-1}. The air defense coordinator must establish the true velocity of this unidentified fighter to determine if it poses an immediate threat to the sector.

Simultaneously, the civil operations wing of the command center is executing a relief drop to assist displaced families isolated by the conflict. A heavy cargo aircraft is flying horizontally at a constant speed of 70 ms^{-1} at a steady altitude of 180 m above

a flat plain. The crew must release a parachute-free container of emergency medical supplies so that it lands precisely at a designated ground collection point marked by ground teams.

Task:

(a) Find the true velocity of the fighter aircraft.
(b) At what horizontal distance from this point must the aircraft release them?

Item 5

Mr. Rusoke plans to take his family on a vacation in December using their holiday caravan. However, before setting off, the caravan requires specialized mechanical repairs. He contacted a local vehicle recovery service in Mukono to tow the caravan from his residence to the repair garage.

The recovery team arrived with a breakdown truck of mass 1200 kg. The caravan has a mass equivalent to two-thirds the mass of the breakdown truck. The engine of the breakdown truck exerts a constant forward driving force of 3000 N. During the towing process, both the breakdown truck and the caravan experience resistive forces due to the roughness of the road surface, which is characterized by a coefficient of kinetic friction, $\mu = 0.12$. The coupling tow bar connecting the breakdown truck to the caravan is designed with a safety threshold and will snap if the tension force exceeds 1250 N.

Shortly after they depart, Mr. Rusoke sets off from his home in his private car to follow the recovery team to the garage. From a stationary start, he accelerates at a constant rate of 2 m s^{-2} for 10 seconds. He maintains the maximum speed attained for the next 5 seconds, and then applies his brakes to decelerate uniformly to a stop at the garage entrance in a time of 2 seconds. *(Take the acceleration due to gravity, $g = 9.8\ m\ s^{-2}$.)*

Tasks

(a) Calculate:

  (i) the acceleration of the towed system, and
  (ii) the tension in the tow bar.

  Use your findings to determine whether the tow bar will safely manage the trip without snapping.

(b) Sketch a velocity–time graph representing Mr. Rusoke's journey. Using your graph, calculate the total distance from Mr. Rusoke's home to the repair garage.

END
GOOD LUCK!!

P425/2
PRINCIPAL MATHS
Paper 2
August 2026
3 hours

MASAKA DIOCESAN EXAMINATIONS BOARD

Uganda Advanced Certificate of Education

JOINT MOCK EXAMINATIONS 2026

PRINCIPAL MATHEMATICS

Paper 2

3 hours

INSTRUCTIONS TO CANDIDATES

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Graph paper is provided.

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Turn Over

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SECTION A

The item in this section is **compulsory**

Item 1

The leadership of Green Valley Community Water Project manages a solar-powered water distribution system that supplies clean water to households, schools, and small businesses. As the population grows, the management committee wants to determine whether the current system can continue meeting demand or whether investment in a larger storage tank and additional solar pumps is necessary.

The committee recently introduced a computerized monitoring system to estimate daily water demand and forecast future consumption patterns. Available records indicate that when the community population was 1,500, 2,000, and 2,500 people, the average daily water consumption was 45,000 litres, 58,000 litres, and 71,000 litres respectively. To assess the reliability of the computerized demand model, the committee will estimate the daily water consumption for a population of 1,800 people and determine the population corresponding to a recorded daily consumption of 85,000 litres. The model will continue to be used if the estimated daily consumption for 1,800 people is less than 55,000 litres and the estimated population corresponding to 85,000 litres exceeds 3,000 people; otherwise, the committee will review and adjust the model before using it for future planning and decision-making.

To assess the storage capacity of a proposed cylindrical water tank, engineers measured its radius as 2.85 m and height as 5.40 m. Due to limitations of the measuring equipment, the recorded dimensions may differ from the actual values by up to $\pm 0.05m$ in the radius and $\pm 0.08m$ in the height. Before construction begins, the committee wishes to determine whether these measurements are sufficiently accurate for the intended design. The measurements will be accepted for construction if the difference between the largest and smallest possible capacities of the tank, based on the stated measurement errors, does not exceed $10m^3$; otherwise, the dimensions will be remeasured using more precise equipment.

As part of a routine assessment of the water supply system, engineers studied the inflow of water into the storage tank during the first four hours after sunrise. The flow rate was found to be described by

$$f(t) = \frac{t}{4 + t^2}, \qquad 0 \leq t \leq 4$$

cubic metres per hour, where (t) represents the time in hours after sunrise. To estimate the total volume of water pumped into the tank during this period, the project engineer used a numerical approximation based on five equally spaced observations of the flow rate. The committee will accept the estimate for planning purposes if its deviation from the actual volume pumped does not exceed 3%; otherwise, a more accurate method will

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em m^3/hr

CamScanner

be required before the estimate is used.

Task

As a student of Mathematics, analyse the available information and advise the committee, with reasons, on the reliability of the computerized demand model, the suitability of the proposed tank measurements for construction, and the accuracy of the estimated volume of water pumped into the storage tank.

SECTION B

*Answer only **two** items from this section, choosing one item from each of the parts **I** and **II**.*

PART I

Item 2

A fruit juice processing company supplies packaged juice to several schools across the country. As demand for its products continues to grow, the company management is committed to maintaining high product quality while meeting customer expectations. As part of a routine quality assurance review, the Operations Manager and the Quality Assurance Officer have been asked to evaluate the effectiveness of the company's production and packaging processes before further expansion.

According to the Quality Assurance Department, each bottle produced has a probability of 0.04 of being defective, independently of all other bottles. The bottles are packed into cartons containing 15 bottles each. Management will maintain the current quality control procedures if the probability that a randomly selected carton contains no defective bottles is at least 0.50, and if both the expected number and the variance of defective bottles per carton are less than 1. Otherwise, additional quality control measures will be introduced.

To meet increasing demand, the company is considering the installation of a second automatic filling machine. Preliminary observations indicate that the time taken by the machine to fill a bottle is equally likely to take any value between 8 and 12 seconds. Management will approve the machine for large-scale production if the probability that a bottle takes between 9 and 11 seconds to fill is at least 0.50. Otherwise, further investigations will be conducted before a decision is made.

The company also monitors the volume of juice filled into each bottle. The volume of juice in a bottle, measured in millilitres, is represented by the continuous random variable V with probability density function

$$f(v) = \begin{cases} k(v - 240), & 240 \leq v \leq 260, \\ 0, & \text{otherwise.} \end{cases}$$

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Although each bottle is labelled as containing 250 ml of juice, management considers a bottle acceptable if it contains between 245 ml and 255 ml of juice. The management will maintain the current filling settings if the probability that a randomly selected bottl is acceptable is at least 0.50; otherwise, the filling machine will be adjusted.

Task

As a Mathematics student, guide the company management on the decisions to make to determine whether the current quality control procedures are adequate, the production process is producing an acceptable number of defective bottles and the new filling machine is efficient enough for large-scale production.

Item 3

The farmers' cooperative group is investigating whether the amount of fertilizer applied to maize fields influences crop yield. Data were collected from ten farmers during the previous growing season.

Fertilizer Applied (kg/acre)	25	20	33	32	40	45	32	52	68	64
Yield (bags/acre)	14	18	25	23	28	25	35	32	39	35

The cooperative intends to use fertiliser application rates to estimate crop yields for future planning if there is evidence of a sufficiently strong positive relationship of at least 0.75 between fertilizer use and crop yield, and if 42 kg of fertilizers applied per acre yield more than 24 bags per acre. Otherwise, field officers will continue collecting actual yield data from farmers.

The cooperative also packages maize flour into bags labelled 10 kg. The masses of the bags produced by one filling machine are normally distributed with mean 10.2 kg and variance 0.16 kg^2. The company will continue using the current machine settings if the probability that the mass of a randomly selected bag lies between 9.9 kg and 10.2 kg is more than 0.90. Otherwise, the machine will be reset.

Past records show that 6% of flour bags are poorly sealed. During a routine inspection, 250 bags are selected at random. The cooperative inspector will recommend acquiring a new sealing machine if the probability that at most 20 bags are poorly sealed is at most 0.3. Otherwise, a new sealing machine will be procured.

Task

As a Mathematics student, assist the farmers' cooperative by analysing the information and making justified decisions on whether there is need to; continue collecting more data or use the available set of data, reset the packaging machine, procure a new sealing machine.

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PART II

Item 4

A grain-processing machine of mass 24 kg is to be transferred from the school agriculture store to the exhibition tent during the annual Science and Innovation Fair. The school administration wants to determine the safest and most efficient method of transportation.

The machine rests on a rectangular platform ABCD. To test whether it can be moved directly by pushing, four students apply forces of 30 N, 24 N, 12 N and 18 N along the sides AB, BC, CD and DA respectively. The direct-pushing method will be approved only if the magnitude of the single force equivalent to the four forces exceeds 15 N and the direction it makes with AB is less than 20°.

The machine is then placed on a trolley of mass 10 kg and pulled by a rope using a force of 160 N inclined at 20° above the horizontal. While moving on a rough floor, the loaded trolley accelerates uniformly from rest to a speed of 5.0 m s^{-1} in 5 seconds. This method will be approved only if the level of resistance between the trolley and the floor does not exceed 0.45.

For a third method, the machine is loaded onto a trolley of mass 12 kg and connected by a light inextensible string to trailer of mass 24 kg. The two connected bodies move on a horizontal rough floor with coefficient of friction 0.20. A trailer exerts horizontal pulling force of 140 N. The administration will approve this method only if the tension in the connecting string does not exceed 45 N, since a greater tension may break the rope and damage the equipment.

Task

As a student of Mathematics, interpret the situation, choose appropriate methods, and determine which transportation method should be recommended for moving the machine safely to the exhibition tent.

Item 5

A school robotics team from a lakeside school has qualified to present its prototype drone at a regional technology exhibition held across a lake. Before approving the trip, the school management requires the team to justify that their transportation and demonstration plans are both safe and efficient.

To transport the drone, the team plans to use a motorboat. The lake current flows due east at a speed of 3 m s^{-1}, while the motorboat can travel at 8 m s^{-1} relative to the water in a direction due north. The school management will approve this method of transport only if the boat's actual speed relative to the ground exceeds 8.5 m s^{-1}.

During the crossing, the team observes a floating marker drifting with the lake current. At one instant, the marker is 120 m directly east of the boat. Since the marker

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could represent floating debris or another vessel, the crossing will be considered safe only if the minimum distance between the boat and the marker remains greater than 60 m throughout their motion.

After arriving at the exhibition, the team demonstrates the drone's delivery system by launching a sensor package from a platform 20 m above the ground. The package leaves the launcher at a speed of 25 m s^{-1}, making an angle of 40° above the horizontal. The exhibition safety committee has designated a landing zone for the package. The demonstration will be approved only if the package lands more than 70 m from the point directly below the launch platform.

Task

As a student of Mathematics, analyse the motion involved, apply suitable concepts of dynamics, and determine whether the transportation and testing arrangements satisfy the required safety conditions.

END

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P425/2
PRINCIPAL
MATHEMATICS
Paper 2
July - August 2026
3 hours

KAMSSA JOINT MOCK EXAMINATIONS

Uganda Advanced Certificate of Education

PRINCIPAL MATHEMATICS

Paper 2

3 hours

Instructions to candidates

- *This examination paper consists of two sections; A and B.*
- *It has five examination items.*
- *Section A consists of one compulsory item.*
- *Section B has two parts, I and II each with two optional items.*
- *Respond to only two items from Section B, choosing one item from each of the parts, I and II.*
- *Respond to three items in all.*
- *Any additional item(s) responded to will not be scored.*
- *Begin each item on a fresh page.*
- *All necessary working must be shown clearly.*
- *Graph paper is provided.*
- *Where necessary, use $g = 9.8ms\text{-}2$.*

Silent, non-programmable scientific calculators and mathematical tables with a list of formulae may be used.

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SECTION A

Answer all items in this section

ITEM 1

In your school, students have been complaining of high temperatures in classrooms and through their student representatives, they demanded that water cooling cans be installed in classrooms so that they can access cool water to quench thirsty caused by the high temperatures.

In response, the school administration has contracted a company to make cooling cans for the school. The cans should be cylindrical with height $H = 3.5 \pm 0.2\ m$ and radius $R = 1.4 \pm 0.1\ m$. The school administration needs to know the least and greatest volume in cubic meters of water the cans can hold.

In testing the cooling properties of the cans, the temperature(°C) of cooling cans measured every 10 minutes were recorded as 82, 70, 56 and 42 with the can's initial temperature being 93°C.

The school welfare teacher argues that with the can's cooling properties, the water in the can will eventually freeze after some time but can't tell how long it will take. He also wants to know the time it takes for the water to cool to a mean temperature of 50°C.

Task

As the company representative at the school, provide the school administration and the welfare teacher with the answers they need.

SECTION B

Part I

Answer only one item from this part

ITEM 2

At Busega Market, a cooperative of farmers delivers thousands of bunches of Matooke daily. The weight of a mature bunch of Matooke varies due to soil quality, rainfall and farming practices.

The cooperative's data analyst collected data on the weight of 50 bunches of Matooke delivered to the market and found out that the weight of the bunches follow a **normal distribution** and the weights he noted for the 50 bunches were as given below;

50	43	39	36	34	32	30	28	26	24
47	42	38	35	33	32	30	28	26	24
46	41	38	25	33	31	29	27	35	23
45	40	27	34	32	31	29	37	25	22
20	25	27	29	31	32	34	36	39	44

Basing on the analyst's data, the market cooperative has set up three pricing and distribution categories based on weight:

- **Standard Grade**: Bunches weighing between 26kg and 38kg (sold to daily walk-by customers).
- **Premium Grade**: Bunches weighing more than 41kg (sold to high-end hotels).
- **Discount Grade**: Bunches weighing less than a certain weight k are considered too small and sold at a low flat rate.

After setting the three pricing and distribution categories, the cooperative manager took **200** bunches of Matooke and wanted to find out the number of bunches that belong to each category

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but found it difficult to find these numbers. Furthermore, he can't find the maximum weight for Discount Grade bunches.

Task

As the Manager's assistant, assist your boss to find the number of bunches of Matooke that belong to each category from his sample and the maximum weight for Discount Grade bunches given that the cooperative decided that the lightest **15%** of all the Matooke bunches should be classified as Discount Grade.

ITEM 3

During the Music, Dance and Drama (MDD) competitions at Kyaddondo S.S 2026, three judges; Judge Amon, Judge Ashraf and Judge Anna adjudicated the competition and among the three Judges, Judge Anna was the Chief Judge.

The organizing committee agreed that judgment for the chief judge and one of the other two judges would be considered to determine the winner of the competitions. The points awarded to the participating parties in eight items by the three judges are given in the table below;

Judge Anna	72	50	50	55	35	38	82	72
Judge Ashraf	60	55	70	50	50	50	73	70
Judge Amon	50	40	62	70	40	48	67	67

The organizing committee has analyzed the results to determine the second judge's results to consider alongside the results of the chief judge but has failed to make the decision.

Before the competitions officially commence, each of the judges has to give opening remarks to the audience. To determine which judge gives remarks first, a simple play game must be played.

A box containing similar marbles, 4 red, 6 blue and 5 green is put before the judges and each judge picks three marbles one at a time from the box without replacement. The judge who picks marbles of the same color is the first to give opening remarks.

Judge Anna, being the chief judge wants to be the one to give opening remarks first but she does not know the chances she has of achieving this.

Task

As the competition official, help the organizing committee to decide which the other judges' results will be considered with those of the chief judge, giving a reason based on valid computations. Also help Judge Anna to know the chances she has in regards to being the first judge to give opening remarks.

Part II

Answer only one item from this part

ITEM 4

John, an engineering student has a soccer ball of mass **400g** and a tool box of **500g**. He wants to investigate about their motion and behavior patterns when connected.

He places the tool box on a rough horizontal table of coefficient of friction **0.2** and ties a light inextensible rope to the box, passes the rope over a smooth fixed pulley at the edge of the table and then ties the ball to the free end of the rope and the ball hangs freely over the edge of the table.

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John tells his younger brother, Mark, that when he releases his connected system from rest, the ball will move downwards. Mark disagrees to what John asserts claiming that the ball is lighter than the tool box so the ball cannot move the tool box and this sparks a heavy debate between the brothers and approach you to settle the debate before them.

Task

(a) Settle the debate between the brothers, clearly stating the rate at which the ball moves (if it actually moves) and the force in the connecting rope.

(b) John also claims that when the ball hits the ground, it bounces back with a speed of $10ms^{-1}$ at an angle of **30°** with reference to the ground but he is not sure of maximum height from the ground the ball bounces back and seeks for your assistance in determining this height. Provide John with the assistance he seeks.

ITEM 5

Mansoor, a new sailor on his **4km** wide home river wants to cross the river to a newly opened landing site that is directly opposite to the point where he wants to set off from. He wants to sail at ***10km per hour*** and the river flows at ***5km per hour***. Mansoor is aware that if he is to achieve this, he must set his boat to an angle to the river bank but lacks the expertise to accurately determine this angle. He also does not know how much time he needs to cross the river.

Mansoor sets of at **9:45am** and realizes that there is a speed boat sailing due East at ***15km per hour***, initially ***3km*** from him in the direction in which he has set off. According to his little expertise in sailing, he establishes that he is likely to collide with this speed boat in the due course. Unfortunately, he can't determine the distance and time when this impact will happen.

Task

As his mentor in sailing, assist Mansoor to determine the angle he must sail at to cross the home river as required and the time it will take him to cross the river, also ascertain whether collision will happen or not. If collision actually happens, at what distance and time will it happen?

END

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P425/2
MATHEMATICS
Paper 2
29 July 2026
3 hours

ENTEBBE JOINT EXAMINATION BUREAU

Uganda Advanced Certificate of Education

MATHEMATICS

Paper 2

3 hours

INSTRUCTIONS TO CANDIDATES:

The paper has **two** Sections*:* **A** *and* **B**. *It consists of* **five** *items.*

Section ***A*** *has* **one** *compulsory item.*

Section ***B*** *has* **two** *Parts* I *and* II, *Respond to* ***one*** *item from each* Part.

Any additional item(s) responded to will **not** *be scored.*

All responses must be written in the answer booklet(s) provided.

Begin each item on a fresh page.

Silent, non-programmable scientific calculators and mathematical tables with a list of formulae may be used.

Graph paper is provided.

Where necessary, use $g = 9.8\ m/s^2$.

SECTION A

This Section is compulsory.

ITEM 1

A civil engineering company is constructing a drainage canal across a valley. During the design process, the engineers collect measurements, estimate the amount of construction material needed, and develop a computer program to solve design equations.

The width of the canal X = 3.7 km, the distance from the starting point to a reference marker Y = 7.0 km and the distance from the marker to the end of the canal Z = 6.25 km are recorded, each rounded to the stated number of decimal places. To assess the possible effect of measurement errors on the design, the engineers calculate the interval within which the exact value of $\frac{X}{Y-Z}$ lies. The engineers will be given a go-ahead assurance only if the estimated interval is within (4 to 6) km.

To estimate the cross-sectional area of earth to be excavated, the engineers model the profile using the function $y = 3^{2x}$ over the interval $0 \leq x \leq 4$. To come up with this area, the engineers use 5 km equal intervals using approximations. The evacuation of the earth will be easy only if this cross-sectional area is less than 40000 km^2.

One of the design equations cannot be solved directly and must be solved iteratively. The engineers decide to use an iterative numerical method with an initial approximation of 1.4 km. The engineering team is required to come up with a diagrammatic representation of the steps involved in solving this problem. The design equation was given by $2x^3 + 5x - 8 = 0$ estimating the root x with an error less than 0.001. The contract will be approved if this root is not beyond 2 km.

Task

(a) With evidence, find out whether the engineers will be given a go-ahead assurance with the contract.

(b) Will the evacuation of the earth be easy?

(c) Will the contract be approved?

SECTION B

Answer only **two** *items from this* Section, *responding to* **one** *item from each* **Part**.

PART I

ITEM 2

A Mathematics and Physics teacher wants to find out whether students who perform well in Mathematics also perform well in Physics. The teacher randomly selects 10 students and records their scores in the two subjects as shown below.

Mathematics (X)	28	20	40	28	21	31	36	29	33	24
Physics (Y)	30	20	40	28	22	35	35	27	31	23

The teacher wants to observe the relationship between the performance of Mathematics and Physics by these students through a diagram to give a correct comment on the performance of the two subjects. Later, the teacher needs to determine the magnitude of this relationship which will help him make conclusions. It is said that if the magnitude of this relationship is greater than 0.7, then Mathematics is vital in the performance of Physics. A student called Stella who scored 32 in Mathematics will not be

good enough in Physics if she scores a mark less than 25 in Physics and therefore advised to go for coaching.

Tasks

(a) Help the teacher to observe the relationship between the performance of the two subjects and make desired comments.

(b) Is Mathematics vital in the performance of Physics?

(c) Will Stella go for coaching or not?

ITEM 3

A food processing company, Fresh Harvest Ltd**,** purchases potatoes from farmers in different districts. The quality of the potatoes and the efficiency of their delivery determine whether the potatoes arrive at the processing plant in good condition.

To improve its operations, the company carries out three investigations. The logistics manager uses rail, air or road to transport potatoes to the processing plant. Historical records show that the chances of using rail, air and road are 0.3, 0.5, and 0.2**,** respectively. The corresponding chances that the shipment arrives at the processing plant on time are 0.2, 0.8 and 0.1, respectively. For improved operations, the chance that the shipment arrives on time must be greater than 0.4.

The marketing department records the number of customer orders received each day. Let the random variable Y represent the number of orders (in tens) received on a randomly selected day. The probability distribution of Y is given below.

Y	5	6	7	8	9
P (Y = y)	a	0.1	a	0.4	0.1

The marketing department was tasked to calculate the value of E (3Y – 8). If the value obtained is above 150, the marketing department will be given an allowance of Shs 1,000,000, but if less than 150, the marketing department gets an allowance of Shs 500,000. The department also wishes to find out how far the number of orders is spread out from their mean value.

During quality inspection, it is found that 25% of the potatoes supplied are rotten. A quality control officer randomly selects 6 potatoes from a large sack for inspection before processing. Quality is assured if only the chance that at least two potatoes selected at random is less than 0.5.

Tasks

(a) Help and find out whether the operations are improved.

(b) Help and find out the allowance which the marketing department will receive.

(c) Assess how far the number of orders is spread out from their mean value.

(d) Will quality be assured?

PART II

ITEM 4

A construction company is transporting building materials to the top of a hill using a simple pulley system. A 6 kg crate is temporarily placed on a rough loading ramp inclined at 30° to the horizontal while workers prepare the lifting equipment. The crate is observed to be just about to slide down the ramp, indicating that it is in limiting equilibrium. The company intends to find the coefficient of

friction between the mass and the plane. The transportation of the building materials will be easy if the coefficient of friction calculated is less than 0.6. If this condition is satisfied, the company uses Shs 35,000 as transport; if not, then the company uses transport as Shs 40,000. The company intends to minimize the cost of transportation.

The workers then connect the crate to a 3 kg counterweight using a light inextensible string passing over a smooth fixed pulley located at the top of the ramp. Once the system is released from rest, the two masses begin to move. The speed of the two masses will be low if the acceleration obtained is less than 5 m/s^2.

Assume that the pulley is frictionless, the string is light and inextensible.

Task

(a) Will the transportation of the building materials be easy?

(b) Will the company minimize the transportation cost?

(c) Assess the speed of the two masses.

ITEM 5

A drone delivery company is testing a new aerial delivery system in an open field. During a trial, a delivery package is projected from a point on a horizontal platform with an initial velocity of 28 m/s at an angle of elevation of 60° above the horizontal. The engineers want to determine the greatest height reached by the package and the time taken to reach that height to ensure safe operation. The delivery system will be effective if the height reached is less than 35 m. The time taken to reach this height should below 3 seconds for the package to be allowed to move in the airspace.

At the same time, two monitoring drones, Drone A and Drone B, are moving through the airspace to track the package. Their positions at time *(t)* seconds are represented by the following position vectors: $r_A(t) = (3i + j + 5k) + t(4i + j - 3k)\text{m}$ and $r_B(t) = (i - 3j + 2k) + t(i + 2j + 2k)\text{m}$.

The engineers need to analyze the motion of the package and the two drones to avoid collision and improve the efficiency of the delivery system. To do this, the engineers need to determine the value of *t* when A and B are closest, and the least distance between A and B. There is efficiency in delivery only if the least distanced obtained is less than 5 m.

Task

(a) Will the delivery system be effective?

(b) Is the package allowed to move in the airspace?

(c) Will there be efficiency in system delivery?

END

MICHEAL TONNY TGMBO

P425/2
PRINCIPLE MATHEMATICS
PAPER 2
JULY/AUGUST 2026
2½ HOURS

RWENZORI REGION ASSESSMENT BOARD

UGANDA ADVANCED CERTIFICATE OF EDUCATION

PRINCIPLE MATHEMATICS

PAPER 2

2 Hours 30 Minutes

INSTRUCTIONS TO CANDIDATES:

- This paper consists of **two** sections **A** and **B**. Section **B** has **two** parts **I** and **II**
- Section **A** is compulsory.
- Attempt two items in section **B**, choosing **one** item from each of the parts **I** and **II**

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SECTION A

This section is compulsory

Item 1: In a certain school, the head-teacher wishes to improve security around the corners of the compound by installing security cameras. He contacted a technician to identify suitable positions for the installation.

Upon mapping, the technician established an initial position and drew a baseline through the compound. He realized that the path across the compound is modelled by the function, $f(x) = 2\cos x - x^2$, where x is the distance in hundreds of meters from initial position along the baseline.

He also noticed that the path intersects the baseline at two points, the most suitable positions where the cameras are to be installed.

However, due to financial constraints, the head-teacher prefers the position on the positive side of the initial position. He needs to determine the exact location where the camera should be installed using an iterative method to avoid miscalculation.

Additionally, the head-teacher discovered that the area between the path and the baseline can accommodate a tent to serve as a consultation area, provided the area is at least 10,000 square meters.

The technician uses a numerical method with 5 sub-intervals along the baseline to ascertain this area.

Task:

a) Justify the exact position where the head-teacher will install his camera

b) Will the area of interest be enough for the consultation tent? Make recommendation basing on the technician

$\int_1^{\bar{2}} 2\cos x - x^2 \, dx$

SECTION B

PART 1

Attempt one item from this section

Item 2: During an inter-university sports tournament, a survey was conducted to gauge the viewing habits of students regarding the two most popular sports; Football and Cricket. The survey results showed the following probabilities:

- 35% watched football but not cricket
- 10% watched cricket but not football
- 40% did not watch either sport

With the next inter-university tournament expected to attract 500 students and assuming viewing habits remain unchanged, the organizing committee is currently facing a challenge. They need to estimate the number of students expected to watch cricket, given that they do not watch football such that if the number exceeds 50, they would prepare a separate court outside the university.

However, a sample of 138 supporters was selected from the previous tournament who watched football. The table below shows the number of supporters and the time taken (in minutes) to enter the stadium, measured from the start of the game until halftime.

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Time (Minutes)	Number of supporters
$0 < x < 10$	10
$10 < x < 15$	35
$15 < x < 20$	65
$20 < x < 30$	107
$30 < x < 35$	123
$35 < x < 45$	138

The committee wishes to know the number of supporters who will arrive before the most likely time. If this number exceeds 50, they intend to offer a free drink to promote the game.

After the tournament, a survey was conducted among the spectators. A sample of 20 spectators was taken. If the probability of finding at most 12 spectators watching football is less than 0.4, the university plans to invest in activities other than football.

Task:

a) Evaluate whether the university will prepare a separate court outside and ascertain whether the committee will offer a free drink to promote football.

b) Justify whether the university is planning to invest in other activities other than football.

Item 3: A school board is reviewing the effectiveness of their mock examinations as a predictor of final A-level success. The director of studies (DOS) has selected a sample data for 11 students from last years performance in mathematics check on the relationship between mock examinations and final A-level grades

Grades in A-level	A	B	B	C	D	E	B	A	D	C	E
Marks in mock exams	76	41	78	59	14	29	61	86	32	64	51

The DOS needs to assess whether mock examination marks provide a true and reliable estimate of students' expected performance in the final A-level exams.

However, a sample of 81 students is taken, the chance of getting a student scoring above the mean mark is 0.4 but board defines good performance if the chance of getting at least half the number of students scoring above the mean mark is greater than 0.5.

Basing on the district mock results, the marks of the whole district were normally distributed with a mean of 45 and standard deviation of 5, the board want to know the average mark for a sample of 100 students who sat for the mock examinations in their school since the chance of getting a student above the average mark is 0.46. if the average mark exceeds 48, the budget for academics will be increased in order to improve on the performance in school.

Task:

a) Justify whether mock examination marks provide a true and reliable estimate of students' expected performance in the final A-level exams

b) Ascertain whether performance was good basing on the board's definition

c) Make a recommendation on whether the Academics budget will be increased.

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PART 2

Attempt one item from this section

Item 4: A civil engineering firm is analyzing the forces acting on a regular two-dimensional steel truss component used in a pedestrian bridge design. The truss structure is shaped like a regular hexagon, ABCDEF, with each side measuring 2m and has a mass of 500kg.

The structure is subjected to eight internal forces due its formation. These forces and their directions indicated by the order of the letters of the truss are as follows; 9N, 5N, 7N, 3N, 1N, 3N, 5N and 4N act in AB, BC, CD, DE, EF, FA, AD and AC where AB and AE are in the x and y planes respectively.

The engineering team needs a complete vector analysis to ensure the truss can withstand the resultant load of 12N.

However, in the process of lifting the truss to the bridge using a crane the rope snapped and the truss fell from a height of 5m vertically on to the rough ground inclined at 25° to the horizontal with a coefficient of friction of 0.3 and slides using an initial speed of $7ms^{-1}$ for a distance of 8m and hits vertical wall. The team would wish to know the speed at which the truss hits the wall.

Engineers assert that for the truss to remain intact, it should not hit the ground with a velocity more than $2.5ms^{-1}$ or else it will be deformed.

Task:

a) Determine whether truss can withstand the resultant load and its direction of action.

b) Justify whether the truss will remain in its form and the speed with which it hits the wall.

Item 5: On a certain island around Lake Victoria, smugglers are loading one tonne of fuel containers on to a boat using a rough wooden ramp. The ramp is 10m long and inclined at 25° to the horizontal with a coefficient of friction of 0.1. Each container with a mass of 100kg is pushed up towards the boat by a horizontal force of 700N. The smugglers are uncertain about how much time it will take to load the containers but they have a strict time window of only 2 minutes to complete their work.

Just after midnight a marine patrol officer was patrolling the waters near Entebbe. At 1:00AM, his spotter noticed a suspicious motorized boat believed to be smuggling fuel and untaxed goods that was 20km due east of their position and heading due N15°E at $15ms^{-1}$. The smugglers were attempting to reach the safety of a protected island by 2:00AM. The officer is not sure of the time it will take to intercept them since the maximum speed of his boat is $20ms^{-1}$.

Task:

a) Justify whether the smugglers will be able to complete their work in time

b) Will the patrol officer intercept the suspicious motorboat in time.

END:.....

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P425/2
MATHEMATICS
PAPER 2
July/August, 2026
3 hours

LUGAZI DIOCESE MOCK EXAMINATIONS BOARD
Uganda Advanced Certificate of Education
PRINCIPAL MATHEMATICS

P425/2

3 Hours

Instructions:

This paper consists of **two** *sections;* A *and* **B**.
It has **five** *examination items.*

Section **A** *has* **one compulsory** *item.*
Section **B** *has* **two parts**; **I** *and* **II**. *Respond to only* **one item** *from* **each part**. *Respond to* **three** *items in all.*

Any additional item(s) responded will **not** *be scored.*

All responses must be written in the answer booklet(s) provided. All responses must be shown clearly.

Responses to each item should start on a fresh page.
Indicate the item number you are responding to, clearly at the top of the page.

Silent, non-programmable scientific calculators and mathematical tables with a list of formulae may be used.
Acceleration due to gravity ($g=9.8ms^{-2}$)

SECTION A (Numerical methods)

(Compulsory item)

ITEM ONE

The district engineering department wants to install a water pump between two points along the bank of a river. They contact the district land surveyor who had prior information from the surveying department that indicated that the river bank stretches from 1km to 2km and bounded by the curve $(y = 1nx)$ and the line $(y = 2 - x)$. The surveyor advised that a graphical approach would enable us get an approximate point to 1decimal place where to locate the water pump, he also advised that there was need to get the exact point using a suitable iterating method where to locate the water pump and if the error between the two points is less than 0.01, then any of the two points can be used for installation of the water pump.

The district engineering department was also facing a problem of knowing the approximate area, the surveyor informed the department that part of the area was under a line $(y = 2 - x)$, and another under the curve $(y = 1nx)$ and by using 0.2km sub interval an approximate area could be got using a suitable approximate method of estimating the area. The surveyor requested that there was need to get the exact area of the error between the two areas was found to be less than 0.05 then any of the two areas can be used to estimate the area to be irrigated by the water pump.

TASK:

As a mathematics student, interpret, give a methodology, and justify the district engineering department.

SCAN AnyScanner

SECTION B: PART I (PROBABILITY AND STATISTICS)

(Attempt one item from this part of the section)

ITEM TWO

An entrepreneur has two business sections one is an agricultural farm producing oranges that are supplied to his juice industry. The entrepreneur had an agricultural expert who had a sample data of oranges collected from the farm measured to the nearest centimetres that is as below

Diameter(cm)	8	9	10	11	12	13	14
Frequency(f)	9	15	21	32	19	13	11

The agricultural expert advised that there was need to find out the average diameter of oranges and advised that it should be more than 10.5 for the farm to supply quality oranges to the juice industry.

The agricultural expert also advised that there was need to find the unbiased estimate of the variance and hence the unbiased estimate of the standard deviation so as to find the 96% confidence interval around the average diameter of oranges and if the interval lied between 11 and 11.5 then there would be no need to apply fertilizers on the farm.

The agricultural expert noted that the smallest 10% of the oranges were rejected for being too small and 10% rejected for being too big. He advised that if the range is between 10.8 and 11 the farm would continue supplying oranges to the industry.

TASK:

As a mathematics student, interpret, give a methodology and informed decision to the entrepreneur on the findings of the agriculture expert.

SCAN AnyScanner

ITEM THREE

The Kampala Capital City Authority (KCCA) is conducting a study on the efficiency of 14 seater taxis operating in Kampala Taxi Parks, during peak hours of traffic jam, the KCCA statician modelled the time (t) in hours a taxi takes to complete a single trip that was found to be as follows:

$$f(t) = \begin{cases} kt: 0 < t < 1 \\ k: 1 < t < 3 \\ k(3-t): 2 < t < 3 \\ 0; otherwise \end{cases}$$

The statician advised KCCA that if the average time of a single trip of travel that exceeds 65 minutes, there will be need to introduce fly overs in areas highly affected by traffic jam. The statician advised the KCCA authorities that there was need to know the standard deviation of the time of travel as it is a basic in data analysis.

When the statician analysed the data, it was approximately normally distributed and there was need to find the middle 80% of the time taken by the taxis to complete a single journey and if the interval from 1.8 to 2.2 hour lies within the range got then the taxis would not get fuel subsidizes that were being planned to be given out by KCCA.

The statician advised the KCCA authorities that there was need to know the 95% confidence interval around the average time taken from a single trip. The statician advised that if the interval does not lie between 1.5 and 2.5 then KCCA would reduce the 14 seater taxis and introduce buses.

TASK:

As a mathematics student, help Kampala City Authority in analysis and justify the traffic jam problems.

SCAN AnyScanner

SECTION B: PART II (mechanics)

(Attempt one item from this part of the section)

ITEM FOUR

A Safari rally organization committee, organized Paris-Dakar rally and focused mainly on the two drivers, one of Mercedes and the other of Red Bull, Paris-Dakar . rally organizers entrusted an engineer to monitor the drivers, he found out that at one time the Red Bull driver was 180km in front of the Mercedes.
The Mercedes Benz driver started in pursuit of the Red Bull driver from rest with an acceleration of $30kmhr^{-2}$ in pursuit of the Red Bull driver moving with a velocity of $100kmhr^{-1}$ and an acceleration of $20kmhr^{-2}$. The organizers wanted to know the time the Mercedes Benz driver took to overtake the Red Bull driver and in turn the time the Red Bull driver took to overtake the Mercedes Benz driver, the engineer advised that for the safari rally to be successful the first time of overtake should not be beyond 2 hours.
The Mercedes Benz driver drove so fast that he went off the road at a valley point of depth 10m and moved horizontally with a speed of $220kmhr^{-1}$. The engineer informed Paris-Dakar organizers that there was a need to know the horizontal distance from the point the car went off the road to a point where it landed on the ground and advised that the distance shouldn't go beyond 50m so as not to have a lot of damage on a car.
On reaching the valley ground, the engineer noted that the Mercedes Benz car of mass 2,000kg moved on a rough horizontal ground with a co-efficient of friction of 0.25 and driving force of 3,500N before it came to rest.
The engineer advised that the distance moved by the Mercedes Benz car should not go beyond 200m since there was a park of wild animals and therefore the Paris Dakar rally organizers needed to know the safety of wild animals.

TASK

As a mathematics student, give a response to the Paris-Dakar Safari rally organizing committee on the problems encountered by the rally and give valid decisions.

@LUDISSEB2026

SCAN AnyScanner

ITEM FIVE

A hotel has a hexagon design in the middle of its hall. It contacts it's the engineer who finds out that it is a regular hexagon design each of side 1m with forces of 5N, 6N, 4N, 7N, 3N and 2N acting along the sides $\overrightarrow{AB}, \overrightarrow{BC}, \overrightarrow{CD}, \overrightarrow{DE}, \overrightarrow{EF}, and\ \overrightarrow{FA}$ respectively. The engineer advised that there was need to know the single force that represents all the 6 forces and if it is less than 15N then the costs would be minimized. He also advised that the direction of the single force is important in the analysis.

The engineer observed that the design can be disconnected from the roof which is at a height of 4m from the ground level and falls vertically downwards. He advised that there is need to know the velocity with which it reaches the ground level and if it is less than $15ms^{-1}$, then there would no loss of energy.

The engineer observed that the regular hexagon design has a mass of 4kg and slides with a driving force of 4N on a rough horizontal floor which has a co-efficient of friction of $\frac{1}{5}$ until it came to rest. There was a wall in the direction of motion at a distance of 10m and would like to know the safety of the wall of the hotel.

TASK

As a mathematics student, interprete, show the methodology and give directed decisions of the findings of the engineer to the hotel management.

END

SCAN AnyScanner

P425/2
Mathematics
Paper 2
July /August 2026
2 ½ hours

KAYUNGA SECONDARY SCHOOLS EXAMINATIONS COMMITTEE (KASSEC)
JOINT MOCK EXAMINATION 2026

Uganda Advanced certificate of education

MATHEMATICS

Paper 2

2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES:

- *This paper consists of* **two** *sections;* **A** and **B** *It has five examination items.*
- *Section* **A** *has* **one** *compulsory item*
- *Section* ***B*** *has two parts;* ***part*** **I** *and* ***part*** **II**. *Answer one item from each part.*
- *Answer a total of* **three** *items.*
- *Any additional item(s) answered will* **not** *be scored.*
- **All** *answers* **must** *be written in the answer booklets/ sheets provided.*
- *Graph paper is provided.*
- *Silent non-programmable scientific calculators and mathematical tables with a list of formulae may be used.*

Turn over

SECTION A

*This item is **compulsory***

Item 1

An aerospace engineer is designing a specialized cooling fin for a satellite component. The profile of the fin is modeled by a complex curve and the engineer needs to determine its cross - sectional area which can later help him to determine its weight and heat dissipation. The top boundary of the fin's cross section is defined by the function

$$f(x) = \frac{10}{x^2 + 2}$$

The width of the fin bases span from $x = 0\ to\ x = 2\ m$ and he hopes to subdivide the interval into four equal sub-intervals. During testing, the same aircraft is used to deliver 20 packages in 45 minutes and another 40 packages in 75 minutes. He is wondering whether to take 60 packages if time does not exceed 105 minutes otherwise, he is to find time before he does the delivery of 60 packages to have the final decision.

The engineer also wishes to derive a formula for estimating the relative error he made in determining the volume of the cylinder of his aeroplane engine, the errors the radius, height and volume being $\Delta r, \Delta h\ and\ \Delta v$ respectively.

Task: Help the engineer make appropriate decisions using mathematical methods.

$u\,dv$

SECTION B

This section has two parts; **I** and **II**

Part I

Answer **one** *item from this part*

Item 2.

The manager of a food processing plant is conducting a dual audit on quality control and employee demographics due to a recent health advisory.

In the packaging department, three workers—Aileen, Barbara, and Cathy—handle the daily batch. Aileen packs 55% of the total volume, Barbara packs 30%, and Cathy packs the remaining 15%. Quality control records show that the probability of a packet containing broken biscuits is 0.7 if packed by Aileen, 0.2 if packed by Barbara, and 0.1 if packed by Cathy. During a routine inspection, the manager randomly selects a packet from the finished batch and finds it contains broken biscuits. He suspects Aileen is responsible.

On the same day, as part of a health-monitoring directive to track demographic vulnerability to an ongoing disease outbreak, the manager records the exact ages of all 40 workers on the factory floor:

19.2, 21.0, 22.5, 24.8, 25.0, 26.3, 27.1, 28.5,
29.0, 30.2, 31.5, 32.8, 33.0, 34.2, 35.5, 37.0,
38.4, 39.0, 40.2, 41.5, 42.1, 43.5, 44.0, 45.8,
46.2, 47.9, 48.0, 49.5, 50.3, 52.0, 53.4, 55.0,
56.7, 58.0, 59.2, 60.5, 62.1, 63.8, 64.0, 65.0

1668-05

Task

(a) Evaluate the manager's suspicion. Based on the evidence of the broken biscuits, how likely is it that Aileen is responsible?

(b) Provide the manager with the average age of his work-force and by graphical analysis an estimate the age threshold below which the youngest 75% of the workforce falls.

Item 3

Umeme, a major electricity distributor, is evaluating the performance of their light bulbs and employee productivity to improve their operations.

15-24
25-34
35-44
45-54
55-64

They have two types of bulbs:

Type A: Expected to last around 1150 hours, with some variation.

Type B: Expected to last around 1900 hours, with some variation.

The company wants to determine:

For Type A bulbs, what percentage are expected to last more than 1200 hours? If this percentage is less than 50%, they plan to replace all Type A bulbs.

For Type B bulbs, out of 1000 units, how many are expected to last less than 1800 hours? If fewer than 500 are expected to fail early, they may decide to stop using Type B bulbs altogether.

Moreover, the company is interested in understanding whether drinking coffee influences the productivity of their engineers. They recorded the daily cups of coffee consumed and the number of lines of code written by two engineers over a month:

X	10	15	20	25	30	35
Y	68	60	55	48	38	32

Umeme will base on a calculation to know if there is a relationship between x and y and if found its exceeding 0.75 at 5% level of significancy then coffee will be served daily

Task:

(a) Decide whether the company should replace Type A bulbs accordingly and calculate how many of the 1000 Type B bulbs will likely fail before 1800 hours. Also, advise if they should continue using this type.

(b) Recommend whether the company should serve coffee daily to improve employee output.

Answer **one** *item in this part*

Item 4

A construction team at a logistics ware house is setting up an automated roof system. A crate of machinery (Body A, m_1=20kg) sits on rough inclined roof tilted at an angle of θ=30° above the horizontal. This crate is connected by a light, inextensible cable passing over a frictionless pulley to a heavy counterweight (Body B, m_2=15kg) that hangs freely vertically.

The coefficient of kinetic friction between Body A and the roof is μ_k=0.15. When the system is released from rest, the hanging counterweight pulls Body A up the incline.

The resultant force acting on each of the two bodies must be at least 140N, the tension in the cable more than 135N and the rate of change of velocity should not be less than 0.5 ms^{-2} if they are to continue with construction work.

A war missile (mass=5000kg) is stuck in a harbor channel. Two towboats tie cables to the front of the missile to pull it into the docks.

- **Towboat A** pulls with a force of F_1=1,500N directed at an angle of 30° North of East.
- **Towboat B** pulls with a force of F_2=1200N directed at an angle of 45° South of East.

Due to the narrow canal, there is a constant water drag (friction) resisting the forward motion with a force of F_{drag} = 400N directly West.

Task: Help the construction team to;

(a) Determine whether the crate – plane goals above are achievable,

(b) discover what single force is required and the direction it should take in order take out the missile from water.

Item 5

The lake authorities have received reports of illegal fishing activities in a remote part of the lake Bunyonyi. To catch the offenders, a police patrol boat (**Boat P**) is dispatched from the main dock in position, **A,** given by $(3i + 8j)$. The patrol boat travels at a steady speed of **20** kmh^{-1} along a straight route heading 45° north of east.

Meanwhile, an illegal fishing boat (**Boat F**) has been spotted at position, **B**, given by $(8i + 6j)$. The fishermen are trying to reach a protected area whose position is **C,**

$(12i + 10j)$. They are moving at a steady speed of **15** kmh^{-1} directly towards point **C**. The police are aiming to intercept the fishing boat before it reaches the protected area to prevent illegal fishing activities.

The authorities need to determine:

- When and where the police boat can intercept the fishing boat.
- Whether the police can catch the fishing boat before it reaches point C.

Tasks

Help the authorities determine the time at which the police boat is closest to the fishing boat during their journeys, whether the police can intercept the fishing boat before it reaches point **C** and if possible, the coordinates where the police boat should meet the fishing boat.

END

P425/2

PRINCIPAL MATHEMATICS

Paper 2

July /August 2026

3 Hours

SECONDARY SCHOOLS JOINT MOCK EXAMINATIONS, 2026

Uganda Advanced Certificate of Education

PRINCIPAL MATHEMATICS

Paper 2

Time: 3 hours

Instructions:

This paper consists of two sections; **A** and **B.**

It has five examination items.

Section A has one compulsory item.

Section B has two parts; I and II. Respond to only one item from each part.

Respond to three items in all.

Any additional item(s) responded will not be scored.

All responses must be written in the answer booklet(s) provided.

All responses must be shown clearly.

Responses to each item should start on a fresh page.

Indicate the item number you are responding to, clearly at the top of the page.

Silent, non-programmable scientific calculators and mathematical tables with a list of formulae may be used.

Acceleration due to gravity ($g=9.8ms^{-2}$)

@2026 Stem Academic Board *Turn Over*

SECTION A (NUMERICAL METHODS)
(COMPULSORY ITEM)
ITEM I

An entrepreneur has a problem of location of a lighting system on his compound. He consults a land surveyor who has a Global Positioning System (GPS) and finds out that the area is bounded by the line $(y = 10 - x)$ and the curve $(y = e^x)$ along the distances from x=2 km to x =3km. The land surveyor advised the entrepreneur that a graphical approach is the most suitable approximate method for location of a lighting system at the point of intersection between the line $(y = 10 - x)$ and curve $(y = e^x)$ along the distances from x=2 to x=3km. The approximate point should be rounded off to 1 decimal place. But he would like to know the exact point using a suitable iterative formula to find the exact point of intersection. If the error between the two values is less than 0.05 then any of the two points can be used for installation of the lighting system.

The entrepreneur is also encountered with a problem of knowing the approximate area, he consults a land surveyor who finds out that the area is enclosed by the line $(y = 10 - x)$, curve $(y = e^x)$ and the distance from 2km to 3km. One area was noted to be under the line $(y = 10 - x)$ and another to be under the curve $(y = e^x)$. He advised to use 0.05 sub – interval for the area under the curve $y = e^x$ and 0.1 subinterval of the area under the line $y = 10 - x$.

There was a need to find the total approximate area using a suitable approximate method. There was also need to find the exact area using a suitable calculus method.

The land surveyor advised that if the approximate area and the exact area have an error less than 0.03, then any of the two areas can be used to estimate the area of the compound.

TASK:

As a student of mathematics, help the architect to give a response to the businessman that has interpretation, methodology and judgment to the businessman.

SECTION B: PART I (PROBABILITY AND STATISTICS)

(Attempt one item from this part of the section)

ITEM TWO

A board of governors (BOG) of a school is planning to the analysis the mock results of principal mathematics P425/2. It requested the Director of Studies (DOS) to provide the scores and their respective number of students that were as indicated in the data below.

Scores	No. of students
0-<5	44
5-<15	81
15-<30	105
30-<50	146
50-<70	98
70-<90	46

The Board of Governors contacted its statistician who advised that there was need to find the average score of their students. If it was found to be less than 45 then they would call an academic assembly. He also advised that there was need of a graphical approach from which to state the modal age group.

The statistician found out that the data provided was normally distributed. He advised that there was need to find out the unbiased estimate of variance and therefore the standard deviation that would be used in estimation. A sample of 40 students was selected among the given ones.

The probability of success is that of scoring at least 50. The statistician advised that there was need to get the probability of at most 7 successful students. If it is found to be less than 0.4, then the DOS will administer a punishment.

The statistician advised the BOG that there is need to find the 98% confidence interval around the average score. If the interval from 35-45 lies within the interval got, then there would be no need to administer remedial tests.

TASK

As a mathematics student, interprete, use a suitable methodology and give a justification to the Board of Governors over the scores and numbers of students provided by the DOS.

ITEM THREE

The academic committee of Parents Teachers Association (PTA) is concerned about the performance of their Biology and Economics class in a post Mock Examination. They contacted the Director of Studies (DOS) who provided the scores as below.

Candidate	A	B	C	D	E	F	G	H	I	J
Biology	45	63	56	61	75	83	73	50	77	70
Economics	90	64	76	70	55	53	62	85	53	62

The PTA presented the data to a data analyst. The data analyst advised that there was need to represent the data graphically to find out the relationship between Biology and Economics.

The data analyst also advised that there was need to calculate the correlation co-efficient. If found to be greater than 0.75, then there would be need to have students offering the two subjects at one sitting.

The data analyst selected three students A, B and C who found out that the probability of passing Biology was $\frac{3}{5}$, $\frac{1}{5}$ and $\frac{2}{5}$ respectively.

He requested that there was need to formulate a probability distribution of passing Biology by the 3 students. Go ahead and find the average number of passing Biology. If it was found to be greater than one, then there would be need to admit more students of Biology, the standard deviation is also important in the analysis.

The data analyst found out that in a sample of ten students above, the probability of passing Biology was found to be $\frac{2}{3}$. He advised that there was no need to find the probability that at most 2 students passed Biology. If it was found to be greater than 0.3, there would be need to reward students for their performance.

TASK

As a mathematics student, analyse, interprete and justify to the PTA over the performance of their students in the two subjects.

SECTION B: PART II (MECHANICS)

(Attempt one item from this part of the section)

ITEM FOUR

A business man wants to install a regular pentagon solar panel structure. The engineer finds out that the regular pentagon solar panel structure is of mass 10kg and length 2m with forces of 5N, 6N, 3N, 2N and 13N acting along the sides AB, BC, CD, DE and EA respectively. The directions of the forces are in the order of the letters.

The engineer took AB as the reference horizontal axis. The engineer advised that there was need to know the resultant force and should not be beyond 18N for the cost of a rectangular solar panel to be affordable. It was also important to know the direction of the resultant force.

The engineer informed the business man that the solar panel was installed at the top of the roof at a distance 4m from the edge of roof that is inclined at 30° to the horizontal. There was need to know velocity with which it reaches the edge in case it is detached from the roof and slides on a rough inclined roof which has the coefficient of reflection of $\frac{1}{4}$. He advised that the velocity should not go beyond $10ms^{-1}$ otherwise there will be loss of energy

The engineer noted that regular pentagon panel was projected at an angle of 30^0 below the horizontal at the edge with initial velocity as from above at the edge of the roof at an angle 30°. The height above the ground to the edge was measured as 4m by the engineer.

There was need to know the horizontal distance from the height to where the rectangular pentagon solar panel hits the ground. He preferred that it does not go beyond 3m since they are usually children who are playing. There is need to know the safety of these children.

TASK

As a mathematics student, analyse and justify the regular pentagon solar panels problems to the business man.

ITEM FIVE

A certain district is affected by heavy floods, it contacted its technical rescue team personnel to deliver relief packages to the affected residents using two drones A and B. At 11:00am the technical rescue team personnel observed that drones A and B were at positions vectors $\begin{pmatrix} 6 \\ -3 \end{pmatrix}$ km and $\begin{pmatrix} 3 \\ 12 \end{pmatrix}$km and their respective velocities were $\begin{pmatrix} -7 \\ 12 \end{pmatrix}$kmh^{-1} and $\begin{pmatrix} -3 \\ -8 \end{pmatrix}$kmh^{-1}. Both drones were flying at the same altitude above the ground. The technical rescue team personnel later suspected that drones A and B could be heading towards the same point. There was need to investigate whether there is a possibility of collision, hence find the position of collision and state the time of collision.

The technical rescue team personnel commanded drone A to change course so as to avoid collision and therefore be able to deliver the relief package. The technical rescue team personnel commands drone A to release package at a height of altitude 50m above the ground level with a horizontal velocity 20ms^{-1}.

There was need to find the velocity and the horizontal distance from the altitude to point of contact with the ground. The technical rescue team personnel advised that the velocity shouldn't exceed 50ms^{-1} for there to be no loss of energy.

The technical rescue team personnel found out that the relief package of mass 50kg slide with a driving force of 100N on a rough horizontal ground with coefficient of friction between the relief package and the ground as 0.25. It moved until it came to rest.

The technical rescue team personnel wanted to find out whether the package slides and reaches the affected residents who are 1600m away from the point of dropping. Its delivered successfully if it does not reach the residents.

TASK:

As a Mathematics student, give a response to the district affected by the heavy floods, the interpretation, methodology and judgment as per the findings of the technical rescue team personnel.

END

WAKISO-KAMPALA TEACHERS' ASSOCIATION (WAKATA)

WAKATA MOCK EXAMINATIONS 2026

Uganda Advanced Certificate of Education

PRINCIPAL MATHEMATICS

Paper 2

WAKATA 2026

2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES:

This paper consists of **two** *sections,* **A** *and* **B**. *It has* **five** *examination items.*

Section **A** *has* **one** *compulsory item.*

Section **B** *has* **two** *parts,* **I** *and* **II**. *Respond to* **one** *item from each part.*

Respond to **three** *examination items in all.*

Any additional item(s) responded to will **not** *be scored.*

All *responses* **must** *be written in the answer booklet(s) provided.*

Where necessary use the value of $g = 9.8ms^{-2}$.

Silent non programmable scientific calculators and mathematical tables with a list of formulae may be used.

Neat work is a must!!

Turn Over

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SECTION A

This section is compulsory.

Item 1.

A small carrier business operating between Kabale and Mbarara uses motorcycles and small vans to transport parcels. The owner introduced a computerized system to estimate parcel charges, monitor fuel usage and determine container capacities. However, he is not sure whether the calculations are reliable enough for business decisions.

Previous records showed that parcels of masses 200g, 400g and 600g were charged Shs 700, Shs 1200 and Shs 3000 respectively. One customer brought a parcel weighing 185g, while another claimed to have Shs 2200, although the parcel mass was not recorded. The owner will keep the computerized charging system if the charge for the 185g parcel is less than 700, and the estimated mass corresponding to Shs 2200 less than 520g; otherwise, the fare estimates will be reviewed before use.

The business also manufactures conical metallic containers for motorcycles. One container has radius of 4.7cm and height of 12.65cm measured with possible errors of ∓ 0.3 *and* $\mp 0.$ respectively. The owner will use the container measurements for production if the difference between the upper and lower limits of the possible volume does not exceed $140cm^3$; otherwise the container dimensions will be re – measured.

In addition, the fuel consumption of one motorcycle while climbing a hilly road near Kabale is represented by the function $\boldsymbol{y = tanx}$ over the interval $\frac{\pi}{6} \leq x \leq \frac{\pi}{3}$. The transport officer uses ordinates to estimate the total fuel used during the climbing session. The officer will use this estimate for planning if the percentage error is less than 5%; otherwise, a more accurate method for determining the fuel consumption will be required.

Task:

Help the owner of the carrier company interpret the information, choose suitable method and make justified decisions about the charging system, container measurements, and fuel consumption estimate.

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SECTION B

PART I

Respond to only **one** *item from this part*

Item 2.

The management committee of a secondary school became concerned about learners' academic performance, staff punctuality and the safety of the school transport system after several complaints from parents and teachers. During a planning meeting, the committee resolved that future decisions should be guided by proper mathematical analysis and evidence rather than assumptions. They therefore wished to study examination performance in order to improve academic standards, evaluate reliability of the staff punctuality arrangements, and assess whether the road commonly used by the school van was safe enough for regular transport activities.

The marks obtained by learners in a mock examination were grouped below;

Marks interval	Number of students
$0 < x \leq 15$	6
$15 < x \leq 30$	8
$30 < x \leq 35$	15
$35 < x \leq 40$	20
$40 < x \leq 55$	10
$55 < x \leq 65$	7
$65 < x \leq 75$	4

The committee wished the director of studies to estimate the median mark using a suitable statistical graph and also use a computational method to determine the mean mark and standard deviation. The committee will maintain the current teaching plan if the mean mark is at least 50; otherwise, weekly tests will be introduced.

The committee also studied the punctuality of a staff member who travels by either route **A** or route **B**. The chance that he uses route is 0.25. if route **A** is used, the that he arrives late is o.6, while for route **B** it is 0.3. The committee will keep the current staff travel arrangement if the chance that the staff member used route **B**, after arriving late, is less than 0.7; otherwise, otherwise a stricter punctuality plan will be introduced.

In addition, accident records on a road used by the school van showed that the monthly number of accidents is a rectangular distribution from 5 to 12 accidents. The committee wished to determine the expected number of accidents in a month and the chance of having more than 7 accidents. The school will continue using the route for regular transport if this chance is less than 0.4; otherwise, an alternative route will be chosen.

Task:

On behalf of the committee interpret the information, help it to make justified decisions about academic support, staff punctuality and safety school transport route.

Item 3.

A beverage company Namanve is testing the reliability of some production and recruitment decisions before introducing a new factory line. The life time X, in years of a machine componer is modeled by the density function below'

$$f(x) = \begin{cases} \beta & ; \ 2 \leq x \leq 3 \\ \beta(x-2); & 3 \leq x \leq 4 \\ 0 & ; \ otherwise \end{cases}$$

Management only will buy a new component if both mean and the median time exceed 3 yea but the standard deviation is less than 1.2. Otherwise, another make of the machine will considered.

The marketing department is studying two independent customer choices: buying mango jui and buying passion fruit juice. The chances of buying both is $\frac{2}{15}$, while those of buying at le one type of juice are $\frac{3}{5}$. A flavor will be retained on the production line if its individual buy probability is at least 0.3, if not, it will be removed.

During recruitment exercise, each of the trainees performs 7 independent machine trails. Fr past records a correct start is twice as likely as an incorrect start. A trainee will proceed for furt training if he/she makes exactly 5 correct starts.

Task:

Help the management to make informed decisions about the process of purchasing a new machine, flavor retention and to determine the qualification chances of a trainee to the next le

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P425/2
PRINCIPAL
MATHEMATICS
Paper 2
2026
3 hours

UGANDA TEACHERS' EXAMINATIONS SCHEME

Uganda Advanced Certificate of Education

JOINT MOCK EXAMINATIONS

PRINCIPAL MATHEMATICS

Paper 2

3 hours

INSTRUCTIONS TO CANDIDATES:

This examination paper consists of two sections; **A** *and* **B**. *It has* **five** *examination items.*

Section **A** *consists of* **one compulsory** *item.*

Section **B** *has two parts;* **I** *and* **II** *each with two optional items. Respond to only* **two** *items from section* **B**. *choosing* **one** *item from* **each** *of the parts.* **I** *and* **II.**

Respond to **three** *items in all.*

Any additional item(s) responded to will **not** *be scored.*

Begin each item on a fresh page.

All *necessary working* **must** *be shown clearly.*

Graph paper is provided.

Where necessary, use $g = 9.8ms^{-2}$.

Silent, non-programmable scientific calculators and mathematical tables with a list of formulae may be used.

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SECTION A

*The item in this section is **compulsory.***

Item 1

A certain distribution company uses trucks to carry its luggage from Kampala to Busia. The company introduced a module to monitor the charges in relation to weights, speed and distance covered and fuel consumption by the vehicle.

The cost of fuel (y in shillings) for distances (x in kilometers) is as shown in the table below.

Distance, x(km)	10	20	30	40
Cost, y(shs)	2800	3600	4400	5200

The manager will hire another driver if the cost of covering 45km exceeds shs. 6000 but if the driver spent shs. 4000, on fuel then the minimum distance to be covered should be at most 30km or else he/she will lose the job.

The route taken by the truck is described by the function $f(x) = x^3-5x+1$, and at every check point, $f(x) = 0$. Unfortunately, the company did not insert this module into their system, yet the only tool left is **graph** and the driver is confused where exactly the vehicle will stop for police checkpoint, and failure to stop at the right spot, the vehicle will be impounded. The driver is required to be careful only in the first 4 kilometers of the journey since it contains the deadliest checkpoints. His friend is a police man who works at one of the checkpoints between 2km and 3km from the company premises and told the driver to consult their technicians at the main station to iterate the graphical value until appropriate numerical distance is obtained, if not the company should be ready to suffer the loss.

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Task.

As a student of mathematics;

(a) Guide the manager on the values he will use in order to hire or maintain the driver.

(b) Help the company on how they can dodge the loss.

SECTION B

Answer only **two** *items from this section, choosing one item from* **each** *of the parts* **I** *and* **II**

PART I

Item 2

One of the prominent Schools in Uganda has organized a visitation day, but for the previous years, parents always complain of traffic jam along the way to the school, since many parents will be moving on that day. The school is planning to meet the parents of candidates but it requires the modal, average and median time such that majority of the parents can attend the meeting.

The table below shows the time taken by parents to travel to school during visitation day

Time (minutes)	<5	<10	<15	<20	<25	<30	<35	<40	<45
Number of parents	28	45	78	143	280	340	375	395	400

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The meeting will be delayed if the interquartile range is less than 10 minutes.

The school is willing to give T-shirts to parents who will arrive in less than 18 minutes, on condition that majority arrive in less than 23 minutes.

On this very day, some parents used personal car (C) while others used buses (B) and taxis (T), with the proportions 40%, 50% and 10% respectively. For those who used personal car, 9% arrived late, and those who used buses, 6% came late and 20% of those who boarded taxi arrived at school late.

Task

As a student of mathematics,

(a) Help the school in planning such that they can have a wonderful visitation day.

(b) Determine the proportion that the vehicle used is a bus given that it arrives late.

Item 3

The director of studies (DOS) in a certain school is complaining why students are performing well in chemistry than in mathematics. He was trying to compare marks of students in both subjects and the table below shows the marks obtained by 10 students.

Mathematics (x)	10	15	20	15	30	35	40	45	50	60
Chemistry (y)	15	20	35	40	35	50	55	40	55	60

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More teachers will be recruited if there is negative correlation between the performance in the above subjects. And extra lessons shall be created if the correlation is insignificant at 5%.

The DOS is wondering what could be the marks obtained in chemistry if a student scores 25 in mathematics.

During printing of report cards, the secretary occasionally makes mistakes. In a batch of 20 cards, 5 are found to have mistake. A sample of 3 batches were at random and examined, the DOS wishes to know the expected number of mistakes and standard deviation made by the secretary.

Task

As a student of mathematics;

Help the Director of Studies clarify on the disparities of marks in the two subjects and the mistakes in order to make an informed decision.

PART II

Item 4

Peter is a very concerned parent, one day, he bought two toy cars, one of mass 1kg and the other of mass 4kg. To his surprise, he found his 5-year-old boy trying to combine the two toys but the strings keep breaking and sometimes the heavier toy would fall on his feet. This always makes the little boy to cry a lot while playing. Peter decided to get for him a rough plane and tried to arrange the set up as follows;

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For safety of the child, he placed the 1kg mass toy on a 4m long rough plane which has a coefficient of friction 0.25, a smooth fixed pulley was placed at the edge of the plane, light inextensible string was connected to the 1kg mass and allowed to pass over the pulley and a 4kg mass which hangs freely at 2.5m above the horizontal ground. Since he wanted the boy to enjoy the movement of the toys, he tilted the plane such that the angle of inclination θ with the horizontal is $\sin^{-1}\left(\frac{3}{5}\right)$. And the system is to be released from rest. In order for peter's son not to cry again, the following conditions must be met,

Condition 1:

Acceleration of the system must be less than $7ms^{-2}$

Condition 2:

The tension in the string must not exceed 15N

Condition 3:

The maximum kinetic energy with which the 4kg mass hits the ground must be less than 70J.

Task

As a mathematics student,

Determine whether the setup peter made will stop his son from crying and justify your responses.

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Item 5

Pirates always smuggle people's goods in Oceans and seas. Due to that, shipmen and fishermen always try to avoid the paths which they like.

One business man always imports his products from India to Mombasa via Indian Ocean. He uses a global positioning system (GPS) to track the path, speed and position of the ships carrying his goods. One day, when he was checking the progress of his goods, he observed that, at **11:30am** a pirate ship is at a place with position vectors $(-6\tilde{i} + 12\tilde{j})\ km$ and is moving with velocity vector $(16\tilde{i} - 4\tilde{j})$ km/h. At 12:00 noon the ship carrying his goods is at a place with position vectors $(12\tilde{i} - 15\tilde{j})\ km$ and is moving with velocity vector $(8\tilde{i} + 16\tilde{j})\ km/h$. Both ships' velocities are constant and the business man is worried that the pirate might destroy his goods if they meet. However, he sent a rescue team which must reach their ship before 1pm.

When the rescue team saw the pirates ship, one of the soldiers in the recue ship fired a bullet with velocity of $320ms^{-1}$ at an angle of 30^0 to the horizontal. But the bullet must pass over a vertical cliff in order to hit the right target and the bullet must be shot when the ships are less than 10000m from each other.

Task

As a student of mathematics;

Determine whether the business man's goods will reach Mombasa safely or not. Justify your response.

END

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P425/2
PRINCIPAL MATHEMATICS
Paper 2
July/Aug
2 ½ hours

NAMIREMBE DIOCESE COUHEIA SECONDARY MOCK EXAMINATIONS 2026

Uganda Advanced Certificate of Education

PRINCIPAL MATHEMATICS

Paper 2

2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES

This paper consists of two (2) Sections A and B.

It has five examination items.

Section A has one item and it's compulsory.

Section B has two Parts, Part I and II with each having two items.

A candidate is to attempt one item from each part of section B.

In total, three items are attempted only.

Any additional item(s) answered will not be scored.

All working must be shown clearly.

In numerical work, take acceleration due to gravity to be 9.8 ms^{-2} unless it is given.

Mathematical tables and silent non-programmable scientific calculators may be used

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SECTION A

(Compulsory)

Item 1

A group of Senior Six students at a renewable energy innovation camp in Kampala are designing a solar-powered water purification system for a rural community. During calibration of the electronic controller, the students discover that one of the sensor equations governing the voltage output of the controller is represented by: $3x^3 - x - 5 = 0$. The engineering software indicates that instability occurs within a narrow operating interval between $x = -1.4$ and $x = -1.2$. To prevent damage to the purifier, the students are required to investigate the behaviour of the controller within this interval and predict the critical operating point as accurately as possible before testing the device. After obtaining an initial estimate from the digital simulator by interpolation, they refine the value using an iterative correction model based on Newton Raphson embedded in the controller's software until a critical operating value is achieved, correct to two decimal places.

As part of the same project, the students are also studying the energy variation in the filtration unit. The accumulated energy transfer over the interval $1 \leq x \leq 2$ is modelled by: $\int_1^2 (x-1) lnx dx$.

Because the school computer laboratory has limited computational power, the students first estimate the quantity numerically using six equal sections, then later compare their estimate with the exact theoretical value obtained analytically. Finally, they evaluate how reliable their approximation method was by determining the greatest possible discrepancy between the two results. Their approximation is considered reliable only if the magnitude of the discrepancy between the two results is less than or equal to 0.003

Task

a) Determine the critical operating value of the purifier.

b) Determine whether the students' approximation of the accumulated energy transfer is reliable State how the reliability of the approximation can be improved.

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SECTION B

Part I

Attempt only one Item from this part

Item 2

The administration of a large girls' secondary school is concerned about the nutritional wellbeing of finalists preparing for national examinations. A health survey is conducted to study the distribution of body masses among the candidates. The cumulative results collected by the school nurse are summarized below:

Mass (Kg)	*<45*	*<50*	*<60*	*<75*	*<80*	*<85*	*<90*	*<100*	*<105*	*<110*
Cumulative frequency	*0*	*9*	*25*	*46*	*63*	*77*	*82*	*89*	*93*	*94*

The head teacher requests the mathematics department to prepare a statistical report that can guide the school feeding programme for the following academic year.

In preparing the report, the learners are expected to organize the information graphically in a form suitable for identifying the most common body mass among the candidates. The school dietician then uses the findings to estimate the average body mass of the students together with a measure showing how widely the masses vary from the average.

The completed report is intended to help the school determine whether the feeding programme is producing balanced nutritional outcomes among the finalists. The feeding programme is considered to be producing balanced nutritional outcomes if the most common body mass among the candidates and the average mass of the finalists lies between 60 Kg – 70 Kg and the standard deviation of mass does not exceed 10Kg.

Task

Make a report to help the school administration determine whether the feeding programme is producing a balanced nutritional outcome among the finalists based on the statistical data.

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Item 3

A road safety organization in Uganda conducted a study involving 30 drivers to investigate the relationship between road accidents and the use of corrective glasses. Out of all the drivers interviewed, 9 had previously been involved in road accidents. Among those accident victims, 5 were found to wear glasses. Further investigations revealed that among drivers who had never experienced an accident, the probability of wearing glasses was $\frac{1}{3}$. The researchers intend to use the information to estimate the likelihood that a randomly selected driver from the group wears glasses. He also intends to determine whether wearing glasses appears common among accident victims and if this chance is less than 45%, the researchers conclude that wearing glasses does not appear common among accident victims.

In a related university health study, researchers analyzing student wellness records discover that the heights of students follow a normal distribution. It is known that only 1% of the students are taller than 200 cm, while 76% have heights between 165 cm and 200 cm. The university medical team wishes to establish the average height and standard deviation of student heights so as to design lecture-room furniture appropriate for most learners. They also need to estimate the percentage of students whose heights fall below 158 cm in order to identify students who may require specially adjusted facilities.

Task

Help the researchers in achieving the objectives of their investigations

Part II

Attempt only one Item from this part

Item 4

During a mountain rescue training exercise, engineers design a transport mechanism that moves medical supplies between two rescue stations built on two smooth mountain slopes intersecting at right angles. One slope rises at 30°to the horizontal

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while the other rises at 60°. A 2 kg emergency package is placed on one slope while a 4 kg package is positioned on the other. The two packages are connected by a light inextensible cable passing over a smooth pulley fixed at the intersection of the slopes. When the system is released from rest, the rescue engineers monitor the motion in order to determine whether the transport system can safely move the supplies without overstretching the cable. To safely move the supplies, the acceleration of the system must be about 4 ms^{-2} and force developed in the connecting cables should not exceed 18N. They therefore investigate both the motion of the packages and the force developed in the connecting cable.

Later, during field testing under rough weather conditions, the slope carrying the 4 kg package develops a small resistive force due to ice accumulation. The engineers repeat the investigation to determine how the additional resistance alters the motion of the system, given that the coefficient of friction between the surface and the 4Kg supply was computed to be 0.25. The system is recommended for use if the change in the motion of the system is less than 1ms^{-2} as compared to the previous value.

In another stage of the rescue exercise, a 50 kg communication device is suspended between two mountain posts by two light strings of lengths 9 m and 12 m attached to points 15 m apart. The two supporting cables are likely to break if their tensions exceed 400N and 300N respectively.

Task

a) Help the engineers to determine whether the transport mechanism safely moves the supplies and also determine whether the system is recommended or not.

b) Help the rescue team determine if the supporting cables can withstand the load safely.

Item 5

A transport company operating in Uganda is training pilots and marine navigators to handle motion under varying wind and current conditions. During one simulation, a car fitted with weather sensors travels at 20 ms^{-1} on a bearing of 040°. At the same

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time, the wind blows from a bearing of 330°at 10 ms^{-1}. The trainee engineers are required to determine how the wind would actually be experienced by the moving driver so that vehicle stability systems can be adjusted correctly.

In a separate navigation exercise beginning at 8:00 a.m., two patrol boats A and B are initially 5.2 km apart, with A due west of B. Boat B travels steadily in the direction $N20°W$at 13 kmh^{-1}, while boat A moves due north at 12 kmh^{-1}. The marine trainees are expected to analyze the motion of one boat relative to the other in order to predict the navigation path observed from boat A. Using this information, they further determine how far apart the boats will be at 8:30 a.m. so as to avoid collision during the exercise.

Meanwhile, at a nearby sports science laboratory, students investigate projectile motion using a launching device. In one experiment, a particle is projected at an angle of 30°with speed 21 ms^{-1} from a point situated 5 m above horizontal ground. The students analyze the motion to predict how far from the launch point the particle will strike the ground.

In another experiment, a boy throws a ball with speed 40 ms^{-1} at an angle of elevation α. During analysis of the flight data, the researchers discover that the possible times of flight corresponding to a horizontal range of 80 m satisfy the equation: $T^4 - 64T^2 + 256 = 0$. **(Hint: Take g = 10ms^{-2})**

Task

a) Help the pilots and marine trainees to determine the solutions to the tasks

b) Help the learners to predict how far from the launch point the particle will strike the ground and to justify mathematically whether the observed flight times satisfy the above relationship.

END

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P425/2

PRINCIPAL MATHEMATICS

PAPER 2

3 hours

JINJA JOINT EXAMINATIONS BOARD

Uganda Advanced Certificate of Education

MOCK EXAMINATIONS – AUGUST, 2026

PRINCIPAL MATHEMATICS

Paper 2

3 HOURS

Instructions to candidates

This examination paper consists of **two** sections: A and B. It has **five** examination items.

SECTION A consists of one compulsory item.

Section B has two parts **I** and **II** each with **two** optional items. Respond to only **two** items from section B, choosing **one** item from each of parts I and II.

Respond to **three** items in all.

Any additional item(s) responded to will **not** be scored.

Begin each item on a fresh page.

All necessary working must be shown clearly.

Graph paper is provided.

Where necessary use g as $9.8ms^{-2}$

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SECTION: A

Item 1

The Uganda National Roads Authority (UNRA) is upgrading a 50km road that connects a trading centre to a lake community. A sino truck transports gravel from a quarry and off loads it at different points along the road according to the condition of the road surface. The following offloading records were made along the route.

At the trading centre 4 tonnes were offloaded. At the swampy area 40km from the community 3.4 tonnes were offloaded. At the beginning of the hill 20km from the centre 2.8 tonnes were offloaded. 30km from the centre 2.2 tonnes were offloaded. At a distance of 10km from the lake community 1.6 tonnes were offloaded and at the lake community 1.0 tonnes were offloaded. The roads supervisor wants to plan the distribution of stones more effectively for the remaining sections of the road. The engineer wanted to offload stones at a distance of 9km from the community and also wanted to offload 2.09tonnes of stones

Engineers also established the relationship between the quantity of stones offloaded, y tonnes, and the distance x in Kilometers as $e^x \cos x + 1 = 0$ for the section $1 \leq x \leq 2$ (kilometers) for the road design in that particular area.

The engineer wanted to perform a dry run to identify actual location to a degree of accuracy of two decimal places.

Task

(a) Advise the supervisor on the quantity of stones to be offloaded and the location for offloading from the centre.

(b) (i) Help the supervisor to establish the road design graphically, and distance where offload was made.

(ii) Advise the engineer on the most accurate location.

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SECTION B

(PART 1)

Item 2

A manufacturing company partnered with a nearby secondary school to improve efficiency in its community outreach programme. During a general cleaning exercise, the time taken by learners to complete an assignment task was recorded.

At the same time, the company monitored the quality of products manufactured by four machines (A, B, C and D). In addition, traffic flow on the road leading to the company was studied to help plan transportation for workers.

The following information was obtained:

the times taken by learners to complete the activity recorded as follows:

Time (seconds)	5-	10-	15-	20-	30-	50-	100-
Number of learners	35	42	58	14	3	1	0

In order to report effectively the company required the average time, the variance from the mean and the most frequent time by the use of an appropriated statistical diagram.

The manufacturing company uses four machines to produce its products. Machine A produces 30% of the daily output, machine B produces 30%, machine C produce 15% and machine D produce 25%.

The percentages of defective products are: Machine A 1%, Machine B 2%, Machine C 3% and Machine D 5%. In order to work on customer satisfaction, the company wanted the probability of product being defective and also that machine A would be used more to produce the products.

Between 8:00am and 9:00am, 40% of the vehicles travelling on the road move from west to east, while the remaining travel from east to west. A random sample of 250 vehicles is observed.

Task.

(a) Help the company to find the statistical measures of time for informed decision making.

(b) If a product selected at random is found to be defective, determine the probability that it was produced by machine A.

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(c) Determine the probability between 77 and 93 vehicles inclusive travel from west to east during the observation.

ITEM 3

A meteorological agency is studying rainfall patterns and student performance.

In a simple model of the weather in July each day is classified as either fine or rainy. The probability that a fine day is followed by a fine day is 0.8. The probability that a rainy day is followed by a fine day is 0.4. The probability that 1 July is fine is 0.75.

The meteorologist wanted to discover whether 2 July would be fine and also 3 July could be fine.

At the same time, the amount of rainfall (in cm) on a randomly selected day is modeled by the continuous random variable X with probability density function

$$f(x) = \begin{cases} 3\alpha x, & 1 \leq x \leq 3 \\ \beta(6 - x), & 3 \leq x \leq 4 \\ 0, & \text{otherwise} \end{cases}$$

where α and β are constants.

To assess academic performance, a random sample of 250 candidates is selected from the district. The sample has a mean examination score of 68.4 and a population variance of 182.25.

Task

(a) (i) Help the meteorologist investigate probability that 2 July will be a fine day then the probability that 3 July is also a fine day.

(ii) Investigate the probability that 3 July is rainy, given 1 July is fine.

(b) Help to construct a cumulative distribution function X, and hence determine the mediar

(c) Investigate the standard error and construct a 98.4% confidence interval for th population mean.

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Part II

ITEM 4

The Uganda Civil Aviation Authority (UCAA) is conducting a search and rescue training exercise on lake. The exercise involves a rescue helicopter, two emergency response vehicles and two rescue motor boats.

During the exercise, the helicopter launches a rescue package from a cliff with an initial speed of 30 ms^{-1} towards a target located 60m horizontally from the point of projection and at the same level. For safety reasons the launch angle must be between 20° to 67°and at the same time, two emergency response vehicles from opposite ends, A and B, of a 20 m straight road. At the same instant one vehicle starts from rest at A and travels towards B with a constant acceleration of 2 ms^{-2} and the other vehicle starts from rest at B and travels towards A with a constant acceleration of 5 ms^{-2}. The intersection occurs for the vehicle to meet.

Meanwhile, two rescue motor boats P and Q begin moving on the lake. Boat P travels at a constant speed of 20 km/h on a bearing of 060° while boat Q travels at a constant speed of 10 km/h. At 2:00 pm the operator on the boat Q observes that boat P is 4km due west of Q. The intention of operator on Q is to pass as close as possible to P.

Task

(a) Investigate whether the rescue was successful.

(b) Investigate the point on the road where the particle would meet from A.

(c) (i) Investigate the course that must be set by boat Q to pass as close as possible to P and the shortest distance.

(ii) Help the operators to know the time when the rescue will take place.

ITEM 5

An engineering team is transporting equipment to a construction site across a river. The equipment is first pulled up a rough inclined plane with an angle of inclination $sin^{-1}\left(\frac{3}{5}\right)$ coefficient of friction $\frac{1}{8}$, between P and the plane, by a light inextensible string passing over a smooth pulley. A 5kg mass on the incline is connected to a hanging load Q of 4kg. Q is initially 2m above the ground. The system is released from rest with the string taut.

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At the top of the incline, four guide cables exert forces of 4N, 5N, 9N and $6\sqrt{2}$ N along EF, EC, ED and AC respectively, where ABCD is a rectangle and AB = 2AD. The engineer wish to replace these four forces with a single equivalent force.

After unloading the equipment, the team crosses a river 50m wide flowing at a constant speed of 3 ms^{-1}. Their boat has a speed of 2.5 ms^{-1} in still water. The boat leaves a point on one bank to reach a point 35m downstream from its starting point. The crew had two possible courses in crossing the river and the best would be where least time is taken

Task

(a) Investigate the acceleration of the system and the time taken for the hanging mass Q to reach the ground and the distance mass P moves up the inclined plane.

(b) Investigate the single force that could replace the four forces.

(c) advise the crew on the course to be taken.

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P425/2
Principle Mathematics
Paper 2
July/Aug, 2026
3 HOURS

PROVINCIAL EXAMINATIONS AGENCY (PEA)
COUHEIA MOCK EXAMINATIONS 2026
Uganda Advanced Certificate Of Education
PRINCIPLE MATHEMATICS
Paper 2
Time: 3 hours

INSTRUCTIONS TO CANDIDATES

- This paper consists of **two** sections; **A** and **B**; with five examination items.
- Answer a total of **three** items.
- Section **A** has **one compulsory** item.
- Section **B** has two parts **I** and **II**, select one item from each part.
- Any additional item(s) answered will **not be scored**.
- All answers must be written in the answer booklets/sheets provided.
- The graph paper is provided.
- Silent non-programmable scientific calculators and mathematical tables with a list of formulae may be used.
- Where possible use $g = 9.8ms^{-2}$

P T O

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SECTION A (Compulsory)

ITEM. 1

In the process of construction of a water reservoir, a team of surveyors measured the depth profile of land at equal intervals of 10 meters across the proposed site and obtained the following systematic measurements in meters; 2.4, 3.1, 3.8, 4.2, 4.0, 3.5 and 2.8. Engineers want to estimate the cross-sectional area of the land before excavation begins.

The engineer further explains that the amount of water expected to be pumped into the reservoir depends on solving the equation $x^3 - 4x - 10 = 0$, where x represents the operating efficiency factor of the pump. A preliminary estimate suggests that the value of x lies between 2 and 3. Since no exact algebraic solution is readily available.

Before final construction begins, the supervisor had to measure the planned rectangular concrete base of the reservoir and record the length as $30m \pm 0.1m$ and width as $30m \pm 0.1m$. The project committee was worried that inaccurate measurements may affect the construction budget and mount of materials required.

Tasks;

As a mathematics student, if given a chance to analytically advise on the above situation, how would you come up with;

a. The appropriate numerical method to find the approximate area of the land available for excavation.
b. A suitable iterative method to obtain the solution to 3 decimal places
c. The range within which the exact area lies and the percentage error in approximating the area

SECTION B Part I
(Choose one item)

ITEM. 2

In a secondary school, the issuance of school uniforms for senior ones depends on their heights. As they proceed in the successive classes their uniform sizes keep changing and new uniforms need to be issued again. It is the school policy that only when the initial mean height is at **least 125cm** and the inter-quartile range of the heights not exceeding **16cm** is when new uniforms will be issued. The heights are as follows;

Height(cm)	< 100	< 110	< 120	< 130	< 140	< 150	< 160	< 170
Cumulative frequency	0	27	85	215	320	370	395	400

Sometimes the head teacher prefers storing the heights of new senior one students in a graphical format to easily establish the **most common** height initially on admission to guide on the amount of material to purchase as students progress in the next classes. Unfortunately he is not a mathematician.

On the other hand, when a student completes S.4 and joins S.5 at the same school, free school uniform sets will be given as a reward under a condition that the relationship between the first 7 best performed subjects in MOCK and UCE is **at least high**. A student obtained the following grades:

MOCK	A	B	A	D	E	D	B	E
UCE	C	B	C	E	E	C	A	D

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Tasks;

As a mathematics student, if consulted, advise the school to establish;

a. Whether new uniforms will or will not be issued.
b. How the head teacher will store the data and also highlight for him the most likely height to enable the change uniform.
c. In your own view, do you think a set of new uniforms will be given to that student joining S.5 with the scores provided? (Justify your responses backed by calculations)

ITEM. 3

In Uganda, the cases of road accidents are increasing because of wreck less crossing and over speeding of pedestrians and drivers respectively. Because of this, traffic lights have been mounted on most busy roads. A case in point is the Kampala northern bypass whereby at a given point the speeds of the cars are normally distributed. Police found out that 95% cars travelled at a speed less than $85kmh^{-1}$ and 10% travelled at a speed less than $55kmh^{-1}$.

On a certain day, an operation by police was carried out on 5 cars to find out how often these accidents occur. They came to realize that the chance of an accident to occur at this point was twice that of an accident not to occur. The O.C traffic instructed that if the chance that at least 3 accidents occurring in a week exceed 0.49, new road humps will be constructed unless otherwise.

In the process, many cars were impounded and each car had its issues depending on the mechanical conditions found on it. These conditions followed a continuous random variable that was uniformly distributed in the interval **a** to **b**. the mechanical conditions for car A had a lower quartile of 5 and an upper quartile of 9. It is a policy by the police to imprison a driver if the mechanical issues of the car lie in the interval between 2 and 12.

Tasks

As a senior six student if consulted, come up with a clear report on the following backed by calculations;

a. The average speed and proportion of cars that exceeded $50kmh^{-1}$
b. Whether new road humps will be constructed or not.
c. Whether it was possible for the driver of car A to be imprisoned or not.

SECTION B Part II
(Choose one item)

ITEM. 4

A trader in kasese who runs a dairy farm find challenges in delivering the milk uphill to where his dairy shop is located. Since he farm is located down at the foothill of Rwenzori Mountain, he was advised to buy a container and fasten 4 ropes on it to enable 4 men to lift it to the dairy uphill. The ropes were fastened in such a way that each was in the direction $N60^0E, 320^0, S20^0E$ *and* 210^0. The men could provide forces in the ropes equivalent to 30N, 20N, 70N and 60N each respectively.

In the long run, this seemed costly and time wasting. He was advised to instead use a pulley placed at the top of the hill driven by an engine. This made him buy a larger tank of mass 20kg to carry milk weighing 100kg and a cable that would pass over the smooth fixed pulley that would connect the tank to another mass of 10kg that could act as a counter weight. The tank containing milk will be moving up the slope of 3 in 5 that is rough with the coefficient of friction 0.4. The counter weight will hang freely vertically downwards on the other side of the hill to control downward motion for refill.

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The pulley should be strong enough to withstand the force imposed on it by the cable for the combined masses uphill and also wishes that when the engine is switched off, the empty tank should be made to slide down with an acceleration of $1.896ms^{-2}$ under influence of a force PN acting downwards parallel to the plane.

Task;
As a mechanics student if consulted, use clear calculations to come up with proper advise on;

a.
   i. A single force and its direction that is equivalent to the system of forces imposed on the container by the four men.
   ii. Whether buying a pulley that could withstand a total net force of 180N for the combined mass uphill would work well for him.

b. The magnitude of force P required in achieving the required acceleration. How safe can this system be kept working efficiently? (Give scientific ways).

ITEM. 5
A game ranger at Murchison falls national park spots a buffalo running along a straight line towards river Nile. The ranger observes it at post A. at the moment of observation (t = 0), the buffalo is at point B, 120m east of post A and it is running due north at a constant speed of $20ms^{-1}$ relative to the ground.

Simultaneously, a rescue drone is launched from the post A to move as close as possible to the buffalo's path. The drone flies at a constant speed of $10ms^{-1}$ relative to the ground maintaining a fixed direction.
At the instant when the drone is $60\sqrt{3}m$ horizontally away from the buffalo, it projects a tranquilizer dart horizontally towards the buffalo. The dart is released from a height 80m above the ground with an initial speed ums^{-1}. To avoid injuring the buffalo, the speed with which the dart hits the buffalo should not exceed $47ms^{-1}$.

When the buffalo was hit, it got shocked and it jumped upwards a distance of 0.45m high. The game ranger wishes to get analysis of the above information mathematically for better reporting.

Tasks;
As a dynamics student, come up with necessary calculation to help the ranger establish;

a. The closest distance reached by the drone and the time it took to be achieved.
b. If the buffalo was injured or not.
c. The initial velocity with which the buffalo jumped.

END

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P425/2
PRINCIPAL MATHEMATICS
Paper 2
July/August 2026
3 hours

WAKISSHA JOINT MOCK EXAMINATIONS

Uganda Advanced Certificate of Education

PRINCIPAL MATHEMATICS

Paper 2

3 hours

INSTRUCTIONS TO CANDIDATES:

This examination paper consists of two sections; A and B. It has five examination items.

Section A consists of one compulsory item.

Section B has two parts, I and II each with two optional items. Respond to only two items from Section B, choosing one item from each of the parts, I and II.

Respond to three items in all.

Any additional item(s) responded to will not be scored.

Begin each item on a fresh page.

All necessary working must be shown clearly.

Graph paper is provided.

Where necessary, use $g = 9.8\ ms^{-2}$.

Silent, non-programmable scientific calculators and mathematical tables with a list of formulae may be used.

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SECTION A

(Compulsory)

Item 1

A group of Senior Six students at a renewable energy innovation camp in Kampala are designing a solar-powered water purification system for a rural community. During calibration of the electronic controller, the students discover that one of the sensor equations governing the voltage output of the controller is represented by: $3x^3 - x - 5 = 0$. The engineering software indicates that instability occurs within a narrow operating interval between $x = -1.4$ and $x = -1.2$. To prevent damage to the purifier, the students are required to investigate the behaviour of the controller within this interval and predict the critical operating point as accurately as possible before testing the device. After obtaining an initial estimate from the digital simulator by interpolation, they refine the value using an iterative correction model based on Newton Raphson embedded in the controller's software until a critical operating value is achieved, correct to two decimal places.

As part of the same project, the students are also studying the energy variation in the filtration unit. The accumulated energy transfer over the interval $1 \leq x \leq 2$ is modelled by: $\int_1^2 (x-1) lnx dx$.

Because the school computer laboratory has limited computational power, the students first estimate the quantity numerically using six equal sections, then later compare their estimate with the exact theoretical value obtained analytically. Finally, they evaluate how reliable their approximation method was by determining the greatest possible discrepancy between the two results. Their approximation is considered reliable only if the magnitude of the discrepancy between the two results is less than or equal to 0.003

Task

a) Determine the critical operating value of the purifier.

b) Determine whether the students' approximation of the accumulated energy transfer is reliable State how the reliability of the approximation can be improved.

SECTION B

Part I

Attempt only one Item from this part

Item 2

The administration of a large girls' secondary school is concerned about the nutritional wellbeing of finalists preparing for national examinations. A health survey is conducted to study the distribution of body masses among the candidates. The cumulative results collected by the school nurse are summarized below:

Mass (Kg)	*<45*	*<50*	*<60*	*<75*	*<80*	*<85*	*<90*	*<100*	*<105*	*<110*
Cumulative frequency	*0*	*9*	*25*	*46*	*63*	*77*	*82*	*89*	*93*	*94*

The head teacher requests the mathematics department to prepare a statistical report that can guide the school feeding programme for the following academic year.

In preparing the report, the learners are expected to organize the information graphically in a form suitable for identifying the most common body mass among the candidates. The school dietician then uses the findings to estimate the average body mass of the students together with a measure showing how widely the masses vary from the average.

The completed report is intended to help the school determine whether the feeding programme is producing balanced nutritional outcomes among the finalists. The feeding programme is considered to be producing balanced nutritional outcomes if the most common body mass among the candidates and the average mass of the finalists lies between 60 Kg – 70 Kg and the standard deviation of mass does not exceed 10Kg.

Task

Make a report to help the school administration determine whether the feeding programme is producing a balanced nutritional outcome among the finalists based on the statistical data.

while the other rises at 60°. A 2 kg emergency package is placed on one slope while a 4 kg package is positioned on the other. The two packages are connected by a light inextensible cable passing over a smooth pulley fixed at the intersection of the slopes. When the system is released from rest, the rescue engineers monitor the motion in order to determine whether the transport system can safely move the supplies without overstretching the cable. To safely move the supplies, the acceleration of the system must be about 4 ms^{-2} and force developed in the connecting cables should not exceed 18N. They therefore investigate both the motion of the packages and the force developed in the connecting cable.

Later, during field testing under rough weather conditions, the slope carrying the 4 kg package develops a small resistive force due to ice accumulation. The engineers repeat the investigation to determine how the additional resistance alters the motion of the system, given that the coefficient of friction between the surface and the 4Kg supply was computed to be 0.25. The system is recommended for use if the change in the motion of the system is less than 1ms^{-2} as compared to the previous value.

In another stage of the rescue exercise, a 50 kg communication device is suspended between two mountain posts by two light strings of lengths 9 m and 12 m attached to points 15 m apart. The two supporting cables are likely to break if their tensions exceed 400N and 300N respectively.

Task

a) Help the engineers to determine whether the transport mechanism safely moves the supplies and also determine whether the system is recommended or not.

b) Help the rescue team determine if the supporting cables can withstand the load safely.

Item 5

A transport company operating in Uganda is training pilots and marine navigators to handle motion under varying wind and current conditions. During one simulation, a car fitted with weather sensors travels at 20 ms^{-1} on a bearing of 040°. At the same

time, the wind blows from a bearing of 330°at 10 ms^{-1}. The trainee engineers are required to determine how the wind would actually be experienced by the moving driver so that vehicle stability systems can be adjusted correctly.

In a separate navigation exercise beginning at 8:00 a.m., two patrol boats A and B are initially 5.2 km apart, with A due west of B. Boat B travels steadily in the direction $N20°W$at 13 kmh^{-1}, while boat A moves due north at 12 kmh^{-1}. The marine trainees are expected to analyze the motion of one boat relative to the other in order to predict the navigation path observed from boat A. Using this information, they further determine how far apart the boats will be at 8:30 a.m. so as to avoid collision during the exercise.

Meanwhile, at a nearby sports science laboratory, students investigate projectile motion using a launching device. In one experiment, a particle is projected at an angle of 30°with speed 21 ms^{-1} from a point situated 5 m above horizontal ground. The students analyze the motion to predict how far from the launch point the particle will strike the ground.

In another experiment, a boy throws a ball with speed 40 ms^{-1} at an angle of elevation α. During analysis of the flight data, the researchers discover that the possible times of flight corresponding to a horizontal range of 80 m satisfy the equation: $T^4 - 64T^2 + 256 = 0$. (Hint: Take $g = 10ms^{-2}$)

Task

a) Help the pilots and marine trainees to determine the solutions to the tasks

b) Help the learners to predict how far from the launch point the particle will strike the ground and to justify mathematically whether the observed flight times satisfy the above relationship.

END

P425/2
Principal Mathematics
Paper 2
July 2026
3 Hours

ACEITEKA JOINT MOCK EXAMINATIONS 2026

Uganda Advanced Certificate of Education

Principal Mathematics

Paper 2

Time: 3 hours

INSTRUCTIONS TO CANDIDATES:

- *This examination paper consists of* **two** *sections;* **A** *and* **B**. *It has* **five** *examination items.*
- *Section* **A** *consists of* **one compulsory** *item.*
- *Section* **B** *has two parts,* **I** *and* **II** *each with two optional items. Respond to only* **two** *items from Section* **B**, *choosing one item from* **each** *of the parts,* **I** *and* **II**.
- *Respond to* **three** *items* **in all**.
- *Any additional item(s) responded to will* **not** *be scored.*
- *Begin each item on a fresh page.*
- **All** *necessary working* **must** *be shown clearly.*
- *Graph paper is provided.*
- *Where necessary, use* $g = 9.8\, ms^{-2}$
- *Silent, non-programmable scientific calculators and mathematical tables with a list of formulae may be used.*

SECTION A:

*Item in this section is **compulsory***

ITEM 1

An engineer is designing a new town road. The layout of the road follows the path $y = 2x + 7cos2x$. This road must cross a straight river located at $y = 0$. To plan bridge construction, the engineer needs to find where the road and river intersect. .The engineer wants to show graphically that the road meets the river at a point between $2km$ *to* $3km$ measured from one boundary B_1 of the town. To pinpoint the exact spot, he plans to use an iterative formula based on the Newton Raphson Method until his answer stabilizes. The difference between two consecutive approximations must be less than a tolerance threshold of 0.0001 km. The town will only approve the project if the determined approximate location of the bridge falls within a safety zone between 2.0440 *km and* 2.0452 km measured from the boundary B_1.

The engineer also plans to first visit the location of the river to make some observations. He plans to use a bus that moves between the boundaries of the town along this road. The bus starts from B_1 and makes stops at different stages 500m apart the bus fares up to the first, second, third and fourth stages are shs11,000, shs15,000, shs18,500 and shs20,000 respectively. The engineer has established that he can use interpolation to approximate the fare to the bridge location which 1800m from B_1. The engineer wishes to spend at most shs42,000 to and from back to B_1 or else he postpones the visit for the following day.

The engineer also wants to use an approximate method to determine the area enclosed by the road and the river which is described $\int(2x + 7\cos 2x)dx$ from $x = 3km$ to $x = 4km$ using 6 ordinates but he has been opposed by some of his colleagues claiming that this will result into an error of more than 1% and this will give wrong conclusions on the costs involved in the contract. The engineer wishes to determine the percentage error in this area to ascertain whether his friends are right so that he can address the fear of his colleagues.

Task:

Help the engineer obtain the information he needs to make informed decisions.

SECTION B:

PART I: ***Respond to one item from this part***

PART I

Item 2

In the bid to improve the performance of mathematics, the head of curriculum in a certain school wants to analyze the performance of students in senior six. He wishes to know the mean and standard deviation. He also wishes to graphically show the distribution of the performance from which he will estimate the most likely mark of a particular test, students had performed as shown in the table below;

Marks	30-<40	40-<50	50-<55	55-<60	60-<70	70-<90	90-<100
Number of students	2	15	19	22	18	17	3

He wishes that if the difference between the mean and the mode is less than the standard deviation, then a facilitation class will be arranged only for the challenged students

He notices that the general performance of students in Mathematics is approximately normally distributed with mean 60.65 and standard deviation of 13.3. He wants to determine whether the percentage of students who scored above 75 is less the percentage of students who scored below 45 so that he can arrange for a retooling seminar for the teachers of the subject.

The probability of picking a football player from the mathematics class is 0.2.if a student is from the football player, the probability that they get D1 is 0.6.for students on the football team,the probability of getting D1 is 0.65. The head of the curriculum wants to find the probability that a math student is a football player given that they received a D1.if this probability is less than 0.5, extra Math lessons will be arranged for the football players.

Task:

Help the head of curriculum to prepare the information he needs for his analysis and make informed decisions.

ITEM 3

From the recently concluded mock examinations, 20 students of a certain school were not satisfied with their scores.

As a concern, the director examinations picked 8 of these students' scripts for a remark to determine the effectiveness of the mock examiners of this subject. The following table gives the scores for the 8 scripts;

Original mark	65	65	70	75	75	80	85	85
New mark	50	55	58	55	58	61	65	65

The director examination wants to verify if the two results correlate by using a suitable statistical diagram. The director examination wishes to write to the mock body to address the challenge in case the correlation in the two results is not significant at 1% level of significance.

The secretary to the director examinations recorded the scores of the twenty students using the original results but failed to clearly show the proportion of those who had scord 85 as follows;

Original mark	65	68	70	72	75	80	83	85
Proportion of students	0.05	0.15	0.1	0.22	0.15	0.05	0.1	a

The director examination wants to know the expectation and variance of these marks and also represent this data graphically for easy identification of the mode to be used in his write up to the mock body. He wishes that the difference in the expectation and the mode is less than the standard deviation to recommend the mock body examination for coming mock examinations.

The director examination also has established that the probability of one of these 20 students scoring a distinction in the final examinations is 0.65. He wants to know the percentage of exactly 15 of students scoring a distinction in their finals. If this percentage is less than 30%, then he will have to arrange for facilitation for the class.

Task:

Help the director of examinations obtain the information he needs to make informed decisions.

PART II:

Respond to one item from this part

ITEM 4

A newly trained pilot flying from airfield A to airfield B faces three distinct challenges: managing stability of a cargo, navigating safely to avoid mid-air collision and executing a precise projectile release.

He plans to fly an aircraft from airfield A to airfield B, carrying a 10 kg cargo attached by an inelastic string. During the flight, a constant horizontal force of 50 N acts on the cargo in the opposite direction of motion, causing the string to displace from the vertical by a certain angle. To ensure a safe flight, the pilot needs to determine this angle of displacement and calculate the tension in the string, knowing that if the tension exceeds 120 N, he will be forced to double the strength of the string.

At 8:00 AM, the pilot sets off from airfield A at a velocity of 110 km/h in a direction of 320°. Airfield B is located 32 km away from airfield A in a direction of 250°. Simultaneously, a second aircraft departs from airfield B, flying at the same altitude with a velocity of 104 km/h in a direction of 030°. To avoid a collision, the pilot wants to determine the shortest distance between the two aircraft during their motion and the exact time it occurs. If this shortest distance is less than 10 km, he will have to adjust his flight altitude.

Once past the point of closest approach, the pilot intends to release the cargo at a point when flying at a height of 250 m above ground and when moving at a velocity of 110 km/h, he must release the cargo so that it undergoes projectile motion and lands precisely at a target point C. The ground distance from the point of release to point C must not exceed 220 m for the cargo to remain within a recoverable distance.

Task

Help the pilot get clear information on this flight and make informed decisions before he sets off.

ITEM 5

The traffic control office is monitoring two motorcycle ambulance riders who are responding to calls from different parts of a town in Eastern Uganda. The town centre roundabout is taken as the origin on a coordinate grid.Distances are measured in kilometres and speeds in kilometres per hour. At exactly 10:00 am, the control screen gives the following information.

For rider A: $\boldsymbol{R_A = \begin{pmatrix} 2 \\ 8 \end{pmatrix} km, V_A = \begin{pmatrix} 18 \\ -12 \end{pmatrix} km/h}$

For rider B: $\boldsymbol{R_B = \begin{pmatrix} 8 \\ 2 \end{pmatrix} km, V_B = \begin{pmatrix} 6 \\ 0 \end{pmatrix} km/h}$

The control officer must decide whether the two riders are on a collision course. The situation is dangerous only if the riders would reach the same position at the same time within 30 minutes if neither changes direction or speed. If the situation is dangerous, the officer must warn the riders.

Later, a community tractor breaks down while carrying sacks of maize. A pickup truck is used to tow it along a rough horizontal road to a safe parking yard. Before towing begins, the local mechanic must check whether the operation is safe. The following data is provided.

Mass of broken

quantity	Value
Mass of broken tractor	1500kg
Mass of pickup truck	1700kg
Driving force of pickup	7200N
Coefficient of friction for tractor	$\boldsymbol{\mu = 0.18}$
Road resistance on pickup	950N
Gravitational acceleration	$\boldsymbol{g = 9.8 ms^{-2}}$
connection	Light inextensible chain

The towing is approved only if the tension in the chain does not exceed 4350 N.

The ambulance riders have completely run out of critical medical supplies and have to fetch them from a nearby trading center. The trading center is separated from the town by a fast flowing part of a river 0.8 km wide flowing at 5 km/h due East. Rider A is selected to pilot a speedboat capable of travelling at a constant speed of 15 km/h to fetch

supplies from the trading center. Rider A departs from port P to reach a port Q, which is located 0.65 km downstream of point P on the opposite bank. After loading the supplies, the rider must return from point Q to point P; however the boat's speed drops to 12 km/h on this return journey. The speedboat's digital navigation display runs entirely on a small battery which, at the start of the journey, indicates only 12 minutes of power remaining before shutdown.

Tasks:

Help to decide whether the:

a) ambulance riders should be warned.

b) towing operation should be approved.

c) rider A will return to point P before the battery shuts down.

END

PAPER
MATHEMATICS
PAPER 2
July/August 2024
3 Hours

[illegible]
Uganda Advanced Certificate of Education
PRINCIPAL MATHEMATICS

PAPER

3 Hours

Instructions:

This paper consists of two *sections: A and B.*
It has five *examination items.*

Section A has one compulsory *item.*
Section B has two parts, *I and II. Respond to only* one item *from each part. Respond to* **three** *items in all.*

Any additional item(s) responded will not be scored.

All responses must be written in the answer booklet(s) provided. All responses must be shown clearly.

Responses to each item should start on a fresh page.
Indicate the item number you are responding to, clearly at the top of the page.

SECTION A (Numerical methods)

(Compulsory item)

ITEM ONE

The district engineering department wants to install a water pump between two points along the bank of a river. They contact the district land surveyor who had prior information from the surveying department that indicated that the river bank stretches from 1km to 2km and bounded by the curve ($y = 1nx$) and the line ($y = 2 - x$). The surveyor advised that a graphical approach would enable us get a approximate point to 1decimal place where to locate the water pump, he also advised that there was need to get the exact point using a suitable iterating method where to locate the water pump and if the error between the two points is less than 0.01, then any of the two points can be used for installation of the water pump.

The district engineering department was also facing a problem of knowing the approximate area, the surveyor informed the department that part of the area was under a line ($y = 2 - x$), and another under the curve ($y = 1nx$) and by using 0.2k sub interval an approximate area could be got using a suitable approximate metho of estimating the area. The surveyor requested that there was need to get the exac area of the error between the two areas was found to be less than 0.05 then any o the two areas can be used to estimate the area to be irrigated by the water pump.

TASK:

As a mathematics student, interpret, give a methodology, and justify the district engineering department.

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ITEM THREE

The Kampala Capital City Authority (KCCA) is conducting a study on the efficiency of 14 seater taxis operating in Kampala Taxi Parks, during peak hours traffic jam, the KCCA statician modelled the time (t) in hours a taxi takes to complete a single trip that was found to be as follows:

$$f(t) = \begin{cases} kt: 0 < t < 1 \\ k: 1 < t < 2 \\ k(3-t): 2 < t < 3 \\ 0; otherwise \end{cases}$$

The statician advised KCCA that if the average time of a single trip of travel that exceeds 65 minutes, there will be need to introduce fly overs in areas highly affected by traffic jam. The statician advised the KCCA authorities that there was need to know the standard deviation of the time of travel as it is a basic in data analysis.

When the statician analysed the data, it was approximately normally distributed and there was need to find the middle 80% of the time taken by the taxis to complete a single journey and if the interval from 1.8 to 2.2 hour lies within the range got then the taxis would not get fuel subsidizes that were being planned to be given out by KCCA.

The statician advised the KCCA authorities that there was need to know the 95% confidence interval around the average time taken from a single trip. The statician advised that if the interval does not lie between 1.5 and 2.5 then KCCA would reduce the 14 seater taxis and introduce buses.

TASK:

As a mathematics student, help Kampala City Authority in analysis and justify the traffic jam problems.

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SECTION B: PART II (mechanics)

(Attempt one item from this part of the section)

ITEM FOUR

A Safari rally organization committee, organized Paris-Dakar rally and focused mainly on the two drivers, one of Mercedes and the other of Red Bull, Paris-Dakar rally organizers entrusted an engineer to monitor the drivers, he found out that at one time the Red Bull driver was 180km in front of the Mercedes.

The Mercedes Benz driver started in pursuit of the Red Bull driver from rest with an acceleration of $30kmhr^{-2}$ in pursuit of the Red Bull driver moving with a velocity of $100kmhr^{-1}$ and an acceleration of $20kmhr^{-2}$. The organizers wanted to know the time the Mercedes Benz driver took to overtake the Red Bull driver and in turn the time the Red Bull driver took to overtake the Mercedes Benz driver, the engineer advised that for the safari rally to be successful the first time of overtake should not be beyond 2 hours.

The Mercedes Benz driver drove so fast that he went off the road at a valley point of depth 10m and moved horizontally with a speed of $220kmhr^{-1}$. The engineer informed Paris-Dakar organizers that there was a need to know the horizontal distance from the point the car went off the road to a point where it landed on the ground and advised that the distance shouldn't go beyond 50m so as not to have a lot of damage on a car.

On reaching the valley ground, the engineer noted that the Mercedes Benz car of mass 2,000kg moved on a rough horizontal ground with a co-efficient of friction of 0.25 and driving force of 3,500N before it came to rest.

The engineer advised that the distance moved by the Mercedes Benz car should not go beyond 200m since there was a park of wild animals and therefore the Paris Dakar rally organizers needed to know the safety of wild animals.

TASK

As a mathematics student, give a response to the Paris-Dakar Safari rally organizing committee on the problems encountered by the rally and give valid decisions.

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ITEM FIVE

A hotel has a hexagon design in the middle of its hall. It contacts it's the engineer who finds out that it is a regular hexagon design each of side 1m with forces of 5N, 6N, 4N, 7N, 3N and 2N acting along the sides $\overrightarrow{AB}, \overrightarrow{BC}, \overrightarrow{CD}, \overrightarrow{DE}, \overrightarrow{EF}, and\ \overrightarrow{FA}$ respectively. The engineer advised that there was need to know the single force that represents all the 6 forces and if it is less than 15N then the costs would be minimized. He also advised that the direction of the single force is important in the analysis.

The engineer observed that the design can be disconnected from the roof which is at a height of 4m from the ground level and falls vertically downwards. He advised that there is need to know the velocity with which it reaches the ground level and if it is less than $15ms^{-1}$, then there would no loss of energy.

The engineer observed that the regular hexagon design has a mass of 4kg and slides with a driving force of 4N on a rough horizontal floor which has a co-efficient of friction of $\frac{1}{5}$ until it came to rest. There was a wall in the direction of motion at a distance of 10m and would like to know the safety of the wall of the hotel.

TASK

As a mathematics student, interprete, show the methodology and give directed decisions of the findings of the engineer to the hotel management.

END

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